

February 2026



**India's Railway EPC, Drinking Water & Waste Water Treatment,
Transmission & Distribution, Power Distribution Panels & Compact
Sub-Stations, and SCADA & Automation Systems Market**

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INDUSTRY OVERVIEW

Unless noted otherwise, the information in this section is derived from "India's Railway EPC, Drinking Water and Waste Water Treatment, Transmission and Distribution, Power Distribution Panels and Compact Sub-Stations and SCADA and Automation Systems Market" (Ken Research Report, February 2026) as well as other publications and industry sources. Neither we nor any other person connected with the Issue has independently verified this information. The information may have been re-classified by us for the purposes of presentation. Industry sources and publications generally state that the information contained therein has been obtained from sources generally believed to be reliable, but that their accuracy, completeness and underlying assumptions are not guaranteed and their reliability cannot be assured. Industry sources and publications are also prepared based on information as of specific dates and may no longer be current or reflect current trends. Industry sources and publications may also base their information on estimates, projections, forecasts and assumptions that may prove to be incorrect.

*The Ken Research Report, February 2026 has been prepared by Ken Research Private Limited at the specific request of our Company. The market research process for the report has been undertaken through **secondary research and the Expert Opinion Methodology**, which involves discussing the status of the market with EPC contractors, railway solution providers, transmission and distribution, automation & SCADA companies, and relevant industry bodies.*

Quantitative market information was sourced from publicly available portals and verified secondary sources, supplemented with expert inputs, and is therefore subject to change based on evolving market and business conditions. Ken Research's estimates and assumptions are based on varying levels of quantitative and qualitative analyses, including industry reports, government publications, and information available in the public domain. Forecasts, estimates, predictions, and other forward-looking statements contained in this report are inherently uncertain because of changes in factors underlying their assumptions, or events or combinations of events that cannot be reasonably foreseen. Actual results and future events could differ materially from such forecasts, estimates, predictions, or such statements.

Ken Research has prepared this study in an independent and objective manner, and it has taken adequate care to ensure its accuracy and completeness. Ken Research believes that this study presents a true and fair view of the India's Railway EPC, Drinking Water and Waste Water Treatment, Transmission and Distribution, Power Distribution Panels and Compact Sub-Stations and SCADA and Automation Systems Market, including key segments, industry structure, and market challenges, within the limitations of secondary information and expert discussions. It does not purport to be exhaustive. This study has been conducted with an "overall industry" perspective, and it may not reflect the performance of any individual company. Ken Research shall not be liable for any loss suffered due to reliance on the information contained in this study. This study should also not be considered as a recommendation to buy or not to buy the shares of any company or companies as mentioned in it or otherwise.

1. EXECUTIVE SUMMARY

India's Infrastructure EPC Market

India's EPC (Engineering, Procurement, and Construction) industry delivers **end-to-end project services** - from design and engineering, procurement of equipment and materials, to civil works, installation, commissioning, and long-term O&M. These services cut across multiple infrastructure sectors such as **transport (roads, railways, airports, ports), energy & power (generation, transmission, distribution), urban development (water supply, sewage, smart cities), and industrial/social infrastructure.**

Within this landscape, this report focuses on 5 priority segments: **India's Railway EPC** (electrification, signalling, station modernisation, O&M) and **India's Drinking Water & Wastewater Treatment Market** (municipal supply and treatment, industrial effluent treatment), **India's Power Distribution Panels and Compact Substation**, **India's Transmission and Distribution** and **India's SCADA and Automation markets.** Each sector is backed by strong government programs and growth opportunities.

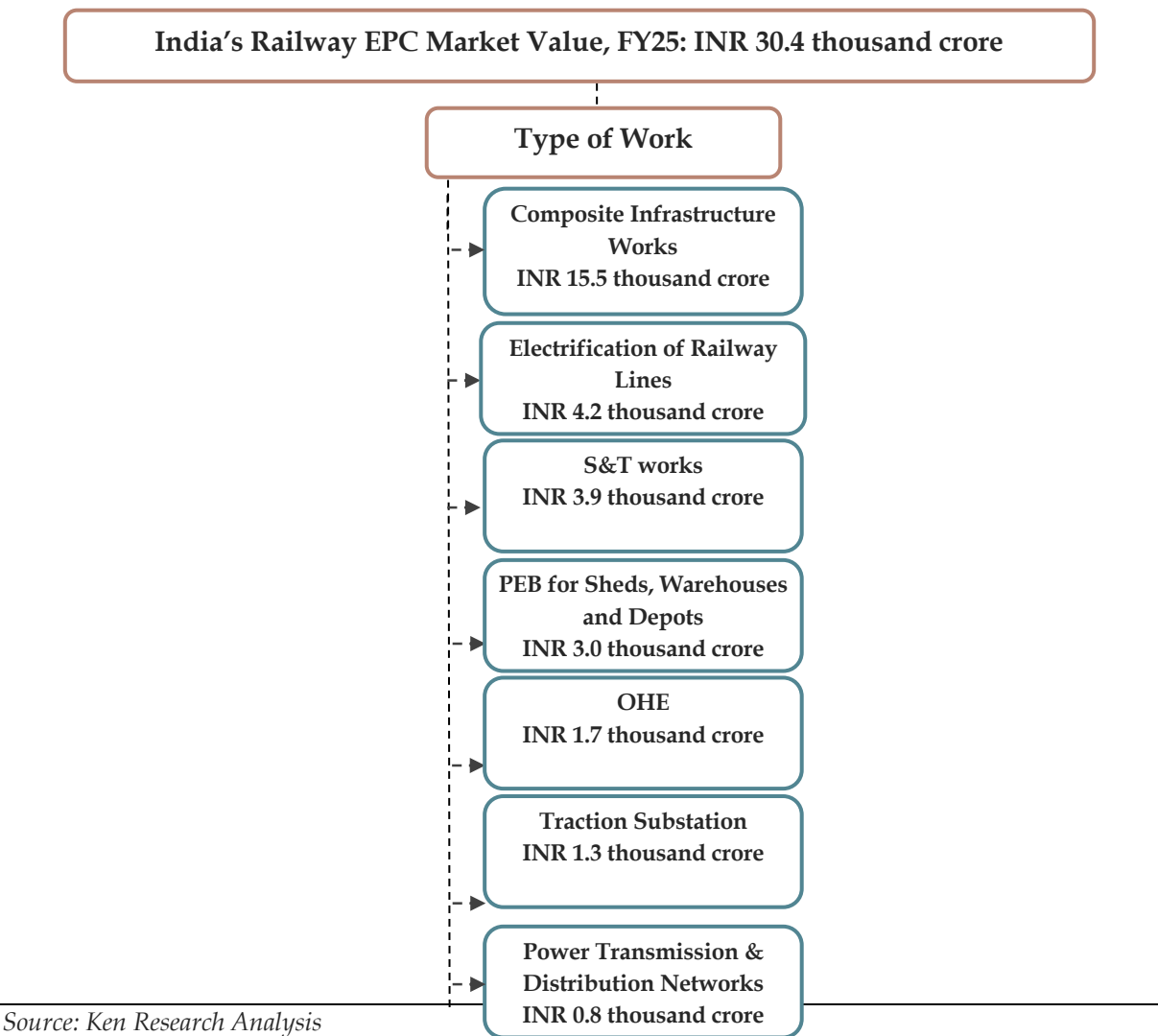
India's Railway EPC Market

The Railway EPC market (excluding track laying works) covers **Civil Engineering Works** (stations, depots, bridges, infrastructure), **Electrical Engineering Works** (overhead electrification, traction substations, SCADA systems), and **Signalling & Telecommunication Works.**

Between FY20 and FY25, the market grew from **INR 16.2 thousand crore** to **INR 30.4 thousand crore**, at a CAGR of **13.4%**, supported by the National Rail Plan, PM Gati Shakti, and investments in station redevelopment, depots, bridges, and electrification. It is projected to rise further to **INR 50.5 thousand crore by FY30F**, growing at a CAGR of **10.7%**, marking a shift from policy-led expansion to

execution-driven demand and sustaining opportunities across civil, electrical, and signalling works. **The split across these works can be seen in the below chart**, with **Composite Infrastructure Works** holding the largest share in FY25, followed by **Electrification of Railway Lines** and **Signalling & Telecommunication (S&T) Works**, while **OHE**, **Power Transmission & Distribution**, **PEB for Sheds, Warehouses & Depots**, and **Traction Substations** account for smaller shares.

Figure 1-1: India’s Railway EPC Market Value (INR thousand crore), FY25

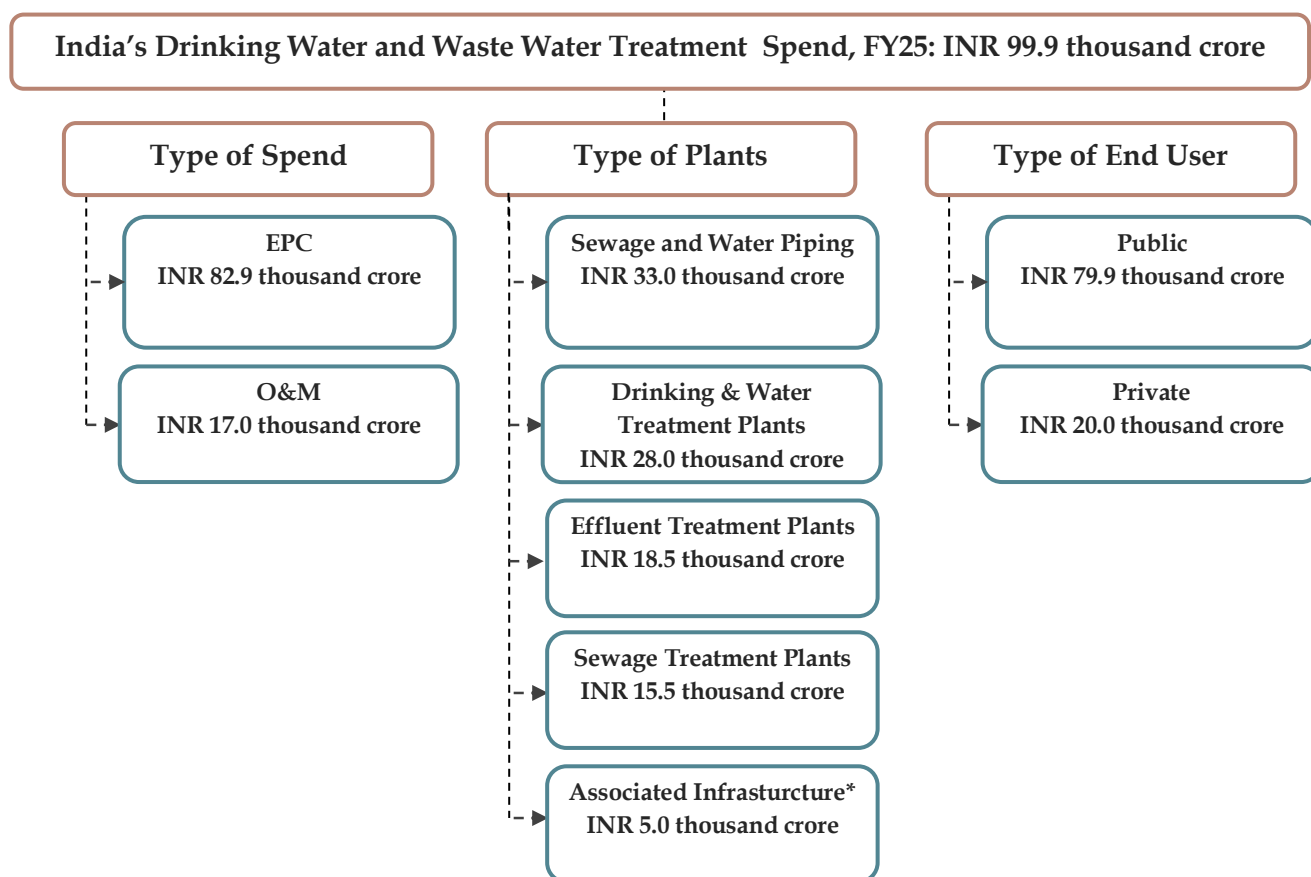


Source: Ken Research Analysis

India’s Drinking Water and Waste Water Treatment
Market

The Drinking Water & Wastewater Treatment market in India covers **DWTPs, STPs, and ETPs**, along with **civil, electromechanical, and SCADA-enabled O&M works** for public and private clients.

Figure 1-2: India's Drinking Water and Waste Water Treatment Market (INR thousand crore), FY25



Source: Ken Research Analysis

Note: *Associated Infrastructure refers to the Balance of Plant which is required for construction and includes pump houses, reservoirs, control rooms etc.

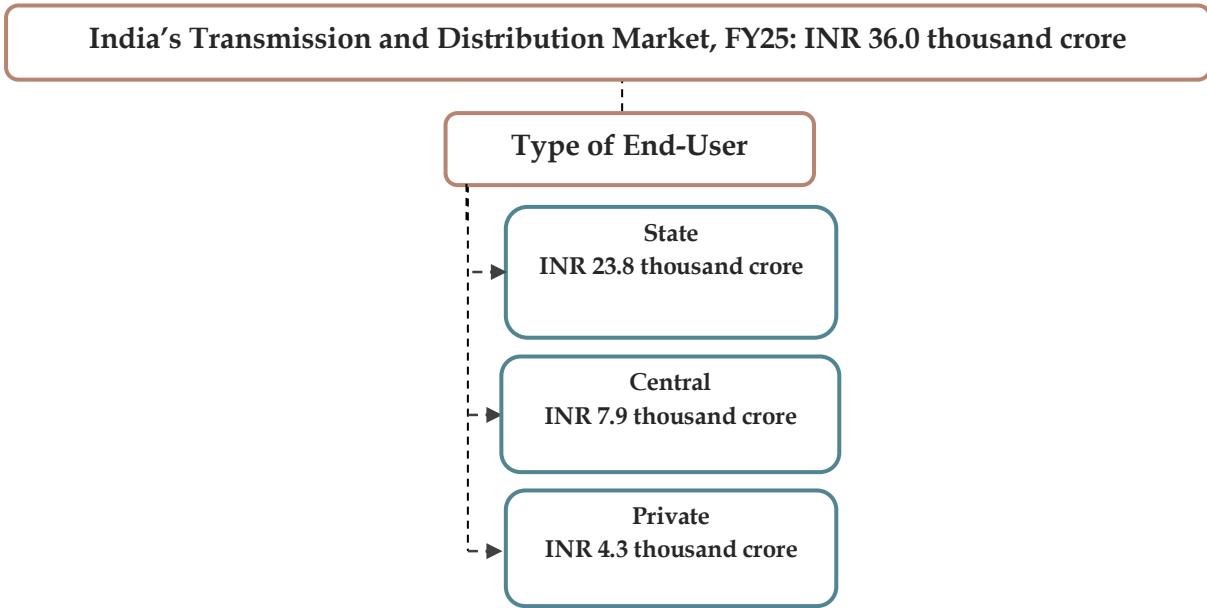
The market grew from **INR 60.5 thousand crore in FY20** to **INR 99.9 thousand crore in FY25 (CAGR 10.6%)**, and is projected to reach **INR 169.6 thousand crore by FY30F (CAGR ~11.2%)**. Growth is supported by government programs but challenged by high capex, operational inefficiencies, and delays in clearances.

Public sector projects dominate spend, led by municipal programs, while **EPC forms the bulk** of activity with **O&M’s share rising** as assets mature. By plant type, **sewage and water networks** lead the market, followed by **DWTPs and ETPs**, while **balance-of-plant infrastructure** is smaller but growing fastest as utilities pursue integrated, reliable systems.

India’s Transmission and Distribution (T&D) Market

India’s T&D market covers **high-voltage transmission lines, substations, feeders, and smart grid systems** connecting generation and consumption centers.

Figure 1-3: India’s Transmission & Distribution Market (INR thousand crore), FY25



Source: Ken Research Analysis

Valued at **INR 36.0 thousand crore in FY25**, the market is projected to reach **INR 60.1 thousand crore by FY30F (CAGR 10.8%)**, driven by renewable integration, **Green Energy Corridors**, and modernization under **RDSS**. **State utilities** remain the largest segment though growth is moderating, while **Central utilities** are expanding rapidly through interregional transmission projects.

Private players are emerging in tariff-based competitive bidding and industrial distribution, marking a shift toward a more diversified, technology-led grid ecosystem.

India's Power Distribution Panels and Compact Substations (CSS) Market

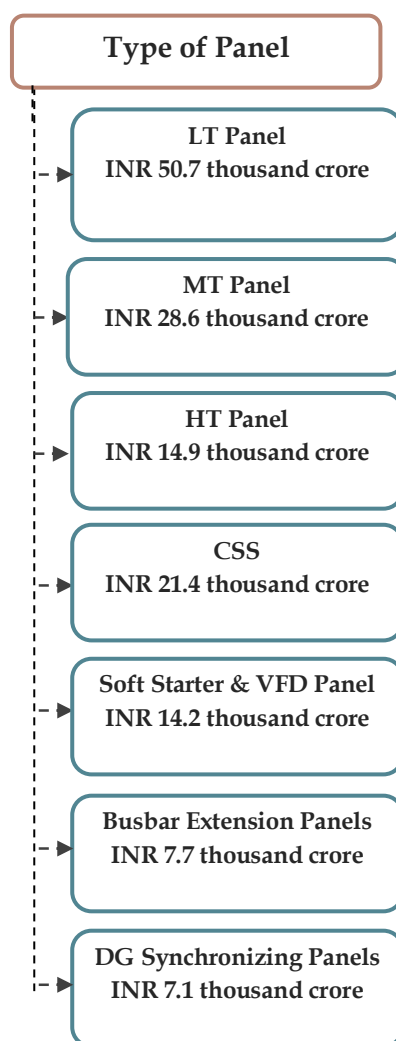
The Power Distribution Panels and Compact Substations (CSS) market in India covers **High Tension (HT, >33 kV)**, **Medium Tension (MT, 1–33 kV)**, and **Low Tension (LT, <1 kV)** panels and substations that enable safe and efficient power distribution across utilities, industries, commercial complexes, and infrastructure projects.

Between **FY20 and FY25**, the market expanded from **INR 105.1 thousand crore** to **INR 144.5 thousand crore**, growing at a **CAGR of 6.6%**, driven by grid modernization, industrial expansion, and electrification programs. It is projected to reach **INR 209.1 thousand crore by FY30F**, registering a **CAGR of 7.7%**, supported by renewable energy evacuation, EV charging infrastructure, and urban distribution upgrades.

Within the market, **Low-Tension (LT) panels** hold the largest share in FY25, followed by **Compact Substations (CSS)** and **High-Tension (HT) panels**, which are also the fastest-growing segment due to state-led grid upgrades and renewable evacuation. Smaller components such as **VFD panels**, **DG synchronization systems**, and **busbar extensions** represent niche but emerging areas, driven by automation and efficiency-focused projects.

Figure 1-4: India's Power Distribution Panels and Compact Substations (CSS) Market (INR thousand crore), FY25

India's Power Distribution Panels and CSS Market, FY25: INR 144.5 thousand crore



Source: Ken Research Analysis

India's SCADA and Automation Systems Market

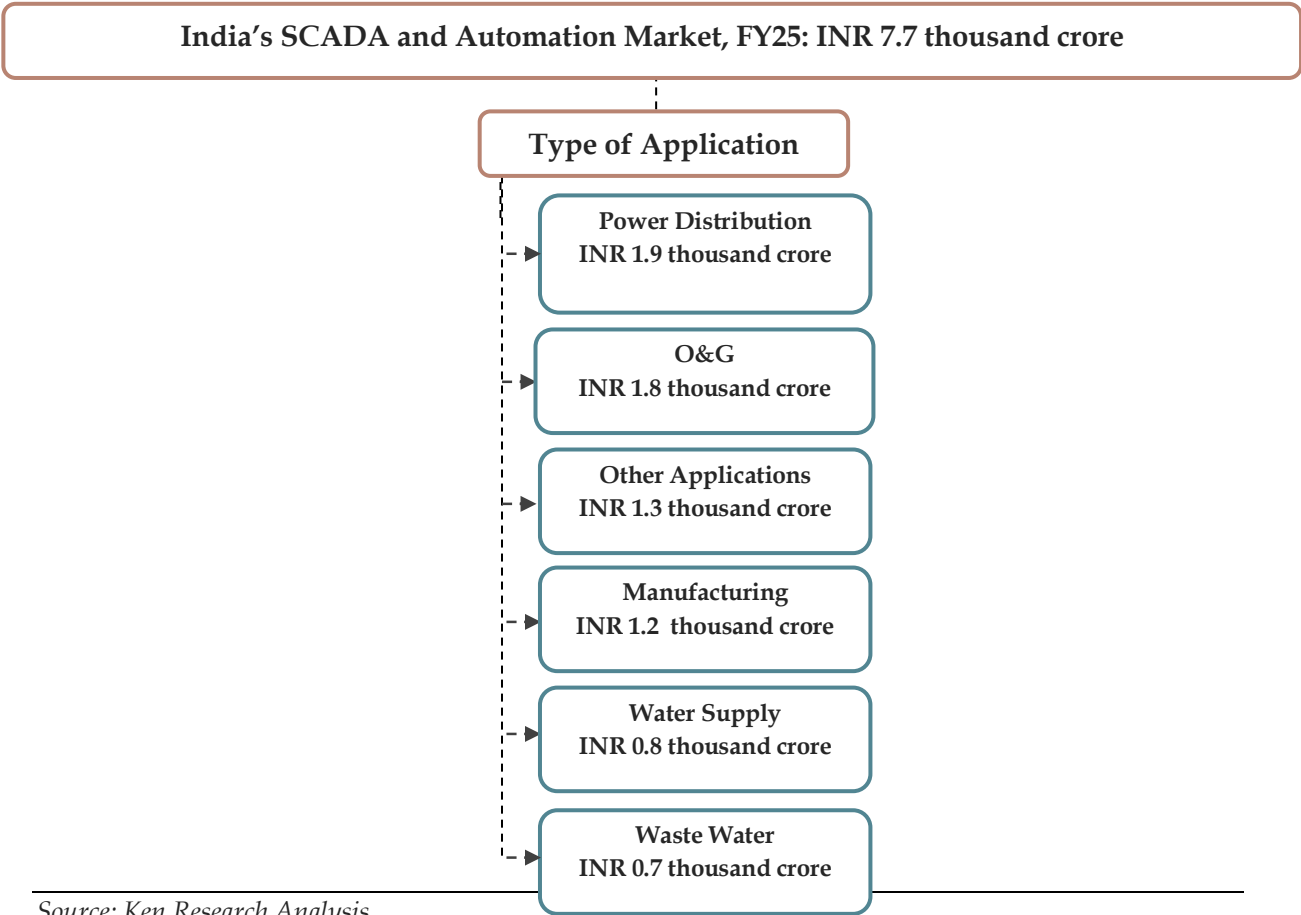
The SCADA and Automation market enables **real-time monitoring, control, and data-driven optimization** across power, water, and industrial systems.

It expanded from **INR 5.1 thousand crore in FY20** to **INR 7.7 thousand crore in FY25** (CAGR 8.6%), and is projected to reach **INR 12.0 thousand crore by FY30F** (CAGR 9.3%), fueled by **infrastructure modernization, renewable integration, and digitalization** across utilities.

Power Distribution SCADA leads the market, while **Water Supply and Wastewater SCADA** are growing

fastest with Jal Jeevan Mission and Smart Cities projects. Industrial SCADA continues to mature, supported by digital upgrades and O&M automation.

Figure 1-5: India’s Power Distribution Panels and Compact Substations (CSS) Market (INR thousand crore), FY25



Source: Ken Research Analysis

The SCADA and Automation market enables real-time monitoring, control, and data-driven optimization across power, water, and industrial systems. It expanded from INR 5.1 thousand crore in FY20 to INR 7.7 thousand crore in FY25 (CAGR 8.6%), and is projected to reach INR 12.0 thousand crore by FY30F (CAGR 9.3%), fueled by infrastructure modernization, renewable integration, and digitalization across utilities. Power Distribution SCADA leads the market, while Water Supply and Wastewater SCADA are growing fastest with

Jal Jeevan Mission and Smart Cities projects. Industrial SCADA continues to mature, supported by digital upgrades and O&M automation.

Conclusion

As projects across these verticals evolve from basic construction toward **technology-driven, high-efficiency operations**, a company performing across all four markets: **Railway EPC, Drinking Water & Wastewater Treatment, Transmission and Distribution, Power Distribution Panels & Substations, and SCADA & Automation** can capture revenue from a **very large and growing market**, leveraging opportunities across both public and private sectors

2. INDIA'S INFRASTRUCTURE EPC MARKET ANALYSIS

Indian Infrastructure EPC Market is defined as the total domestic revenue generated from engineering, procurement, and construction (EPC) activities undertaken within India across Railway Infrastructure and Water Infrastructure segments, along with the related markets for HT/LT Power Distribution Panels, Compact Substations (CSS), and SCADA & Automation Systems. The market is measured in value terms (INR crore), reflects only in-country EPC execution and export revenues are excluded.

2.1. OVERVIEW OF PUBLIC AND PRIVATE INFRASTRUCTURE SPENDING IN INDIA

India's infrastructure buildout is one of the largest globally, with public investments leading the way. Between CY17 and CY23, Infrastructure investments about INR 67 trillion took place in India, and over INR 143 trillion is expected between CY24 and CY30. This highlights more than a two-fold increase (Source: CRISIL). **Government agencies, both central and state, account for nearly 80% of this expenditure**, underscoring the state's dominant role in funding and executing projects.

At the same time, **private sector participation has remained a consistent and growing pillar of India's infrastructure strategy**, particularly through PPPAC-approved projects. These projects, cleared by the Public Private Partnership Appraisal Committee and which are eligible for central support, reflect the government's commitment to structured private involvement.

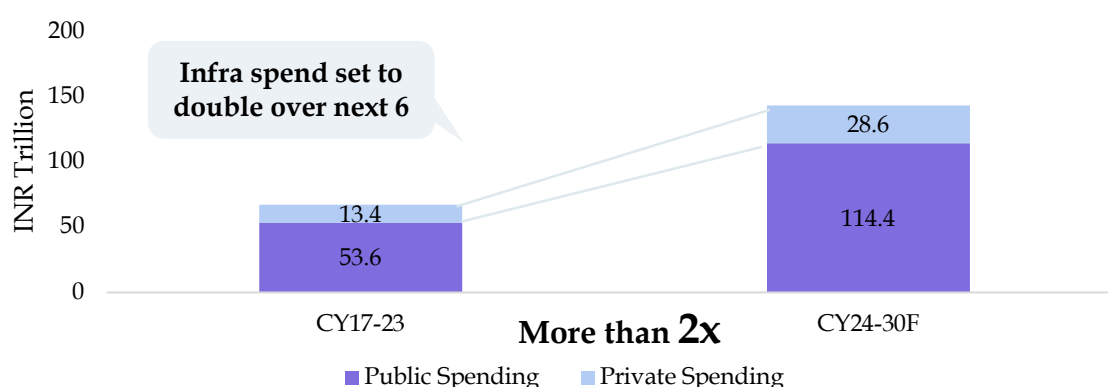
Public and Private expenditure combined provide a strong push for infrastructure development driven by the Government of India is creating **significant opportunities**.

In railways, new lines, station redevelopment, and freight corridors are increasing requirements for EPC works and execution capacity. Wastewater treatment is becoming integral as urban growth and regulatory standards mandate treatment and reuse systems.

In the power sector, transmission expansion and distribution upgrades are driving the use of HT/LT distribution panels and compact substations to support reliable delivery.

Across power, transport, and utilities, the **shift toward digital monitoring and grid stability makes SCADA and automation systems necessary** for real-time control, compliance, and operational efficiency.

Figure 2-1: India's Infrastructure Spend in terms of Value (in INR Trillion), CY17-CY23 & CY24-CY30F



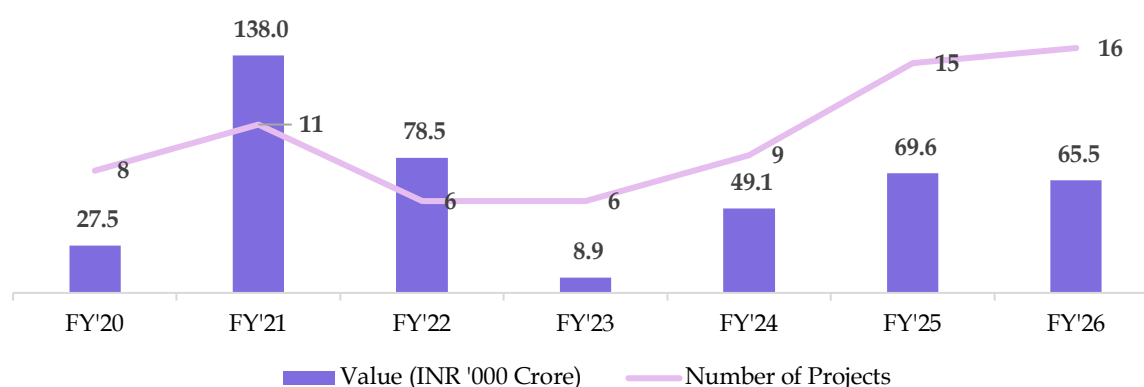
Source: 2023 Infrastructure Yearbook of CRISIL Limited, Ken Research Analysis

Note 1: CY refers to Calendar Year Jan- Dec

The evolving trend of PPPAC-approved projects (project cleared by the Public Private Partnership Appraisal Committee, making it eligible for central support) from

FY20 to FY26 reflects the government's sustained commitment to infrastructure expansion through private sector participation. Notably, FY21 marked the highest project cost at INR 138.0 thousand crore, while substantial approvals continued in FY25 (INR 69.6 thousand crore) and FY24 (INR 49.1 thousand crore). **In FY26**, project approvals will remain steady at INR 65.5 thousand crore, indicating continued momentum in private-sector-driven infrastructure development. Despite fluctuations in the number of proposals each year, the consistent capital allocation underscores a forward-looking national infrastructure strategy.

Figure 2-2: India's PPP Projects by Value (in INR '000 Crores) and Number of Projects (in Absolute Numbers), FY20-FY26



Source: Department of Economic Affairs & Ken Research Analysis

Sector wise Development for Infrastructure Projects

Among the sectors approved for development under PPP frameworks, Some areas offer strong alignment with owing to the large number of projects being executed and their high project value: railway, electricity transmission, electricity generation and water treatment plants.

Table 2-1: Major Sector wise PPP-Projects in India by Value (in INR Crore) and No. of Projects (in Absolute Numbers), FY07-FY26

S.no	Sector	No. of Projects	Total Project Cost (INR Cr.)
1.	Railway track, tunnel, viaducts, bridges	568	5,12,621
2.	Electricity transmission	1,378	4,32,594
3.	Electricity generation (grid)	371	16,30,471
4.	Electricity Distribution	16	28,757
5.	Water treatment plants	45	7,268

Source: Department of Economic Affairs & Ken Research Analysis

Between FY'07 and FY'26, the railway sector accounted for over INR 5.1 lakh crore in projects spanning tracks, tunnels, bridges, and viaducts, generating large-scale requirements in civil construction, electrification, and allied EPC works.

During the same period, the water and wastewater segment saw continued investments in treatment plants, sewage networks, and distribution systems, sustaining steady EPC demand in process engineering and long-term O&M services.

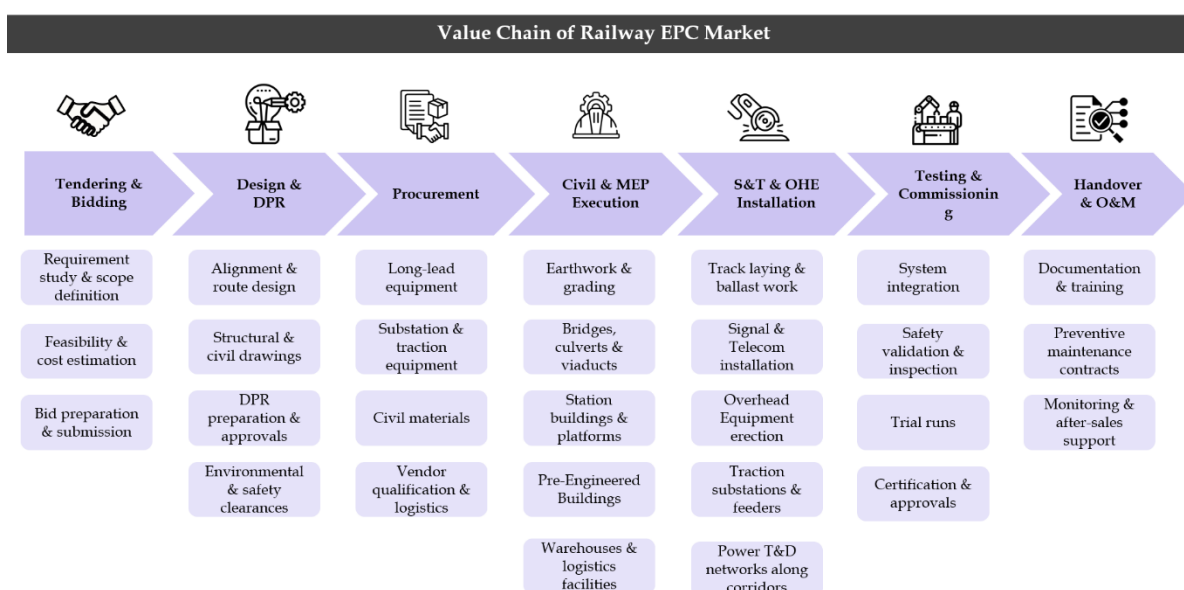
In the power sector, projects worth around INR 4.3 lakh crore in transmission and INR 28,757 crore in distribution required EPC contractors to deliver modern grid infrastructure, with SCADA and automation systems becoming central to project execution. These developments collectively reflected a robust EPC pipeline anchored by strong project flow in rail infrastructure, expanding opportunities in water and wastewater, and an increasing emphasis on automation, substations, and HT/LT electrical integration across power projects.

3. VALUE CHAIN ACROSS TARGET MARKETS

3.1. VALUE CHAIN OF INDIA'S RAILWAY EPC MARKET

The value chain of India's **Railway EPC projects** initiates from Tendering & Bidding of EPC projects. Once the project is commercialised the subsequent steps are initiated which include Design & Detailed Project Report (DPR), Procurement, Civil & Mechanical, Electrical, and Plumbing (MEP) Execution, and Signalling and Telecommunication (S&T) & Overhead Electrification (OHE) Installation, followed by Testing & Commissioning and Handover & Operation and Maintenance (O&M).

Figure 3-1: Value Chain of India's Railway EPC Market



Source: Ken Research Analysis

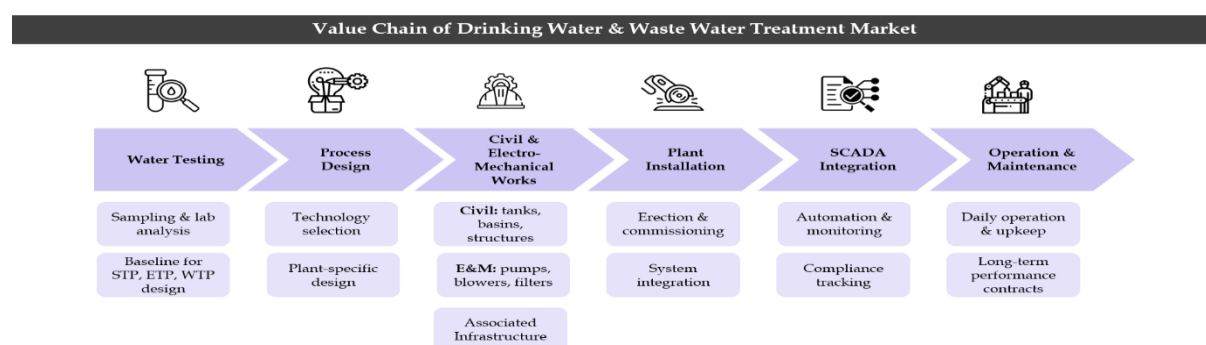
Note: EPC refers to Engineering, Procurement, and Construction, DPR refers to Detailed Project Report, MEP refers to Mechanical, Electrical, and Plumbing, S&T refers to Signaling and Telecommunication, OHE refers to Overhead Electrification and O&M refers to Operation and Maintenance

3.2. VALUE CHAIN OF INDIA'S DRINKING WATER & WASTE WATER TREATMENT MARKET

The value chain of **India's Drinking Water and Waste Water Treatment Plant projects** typically:

- Initiates from Water Testing, which includes sampling, laboratory analysis, and baseline studies for STP, ETP, and WTP projects.
- Once the initial assessments are complete, the process moves to Process Design involving technology selection and plant-specific engineering.
- This is followed by Civil & Electro-Mechanical Works, covering construction of civil structures, tanks, basins, and installation of pumps, blowers, filters, and related infrastructure.
- Subsequent stages include Plant Installation with erection, commissioning, and system integration, SCADA Integration for automation and monitoring.
- Finally, Operation & Maintenance through daily operations, upkeep, and long-term performance contracts.

Figure 3-2: Value Chain of India's Drinking Water & Waste Water Treatment Market



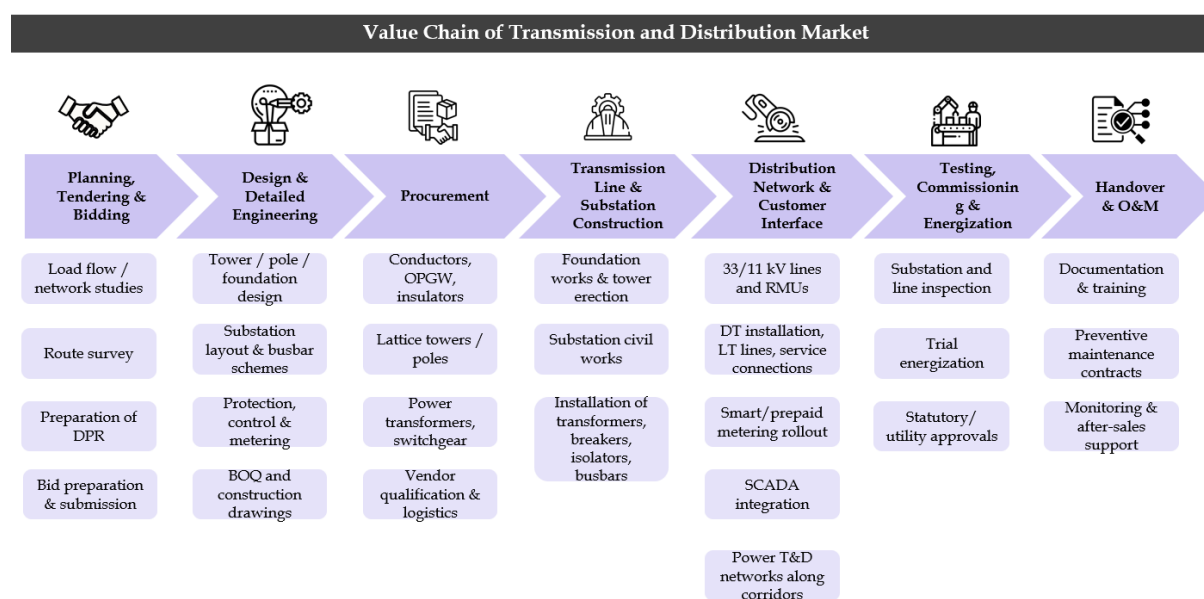
Source: Ken Research Analysis

Note: STP refers to Sewage Treatment Plant, ETP refers to Effluent Treatment Plant, WTP refers to Water Treatment Plant, E&M refers to Electro-Mechanical, SCADA refers to Supervisory Control and Data Acquisition.

3.3. VALUE CHAIN OF INDIA'S TRANSMISSION & DISTRIBUTION MARKET

The value chain of India's Transmission and Distribution (T&D) projects typically starts from network planning, route/scheme finalization, and tendering of EPC packages by central utilities (POWERGRID), state transmission utilities (STUs), discoms, and large industrial/renewable developers. Once the project is awarded, subsequent stages include Detailed Engineering & Tower/Foundation Designs, Procurement of conductors, towers, transformers, switchgear and protection systems, followed by Construction & Erection of transmission lines and substations, Distribution network rollout (11 kV/33 kV), and Metering & SCADA integration. The lifecycle is completed with Testing & Commissioning, Energization, and finally Handover for Operation & Maintenance (O&M) to the utility.

Figure 3-3: Value Chain of India's Transmission & Distribution Market

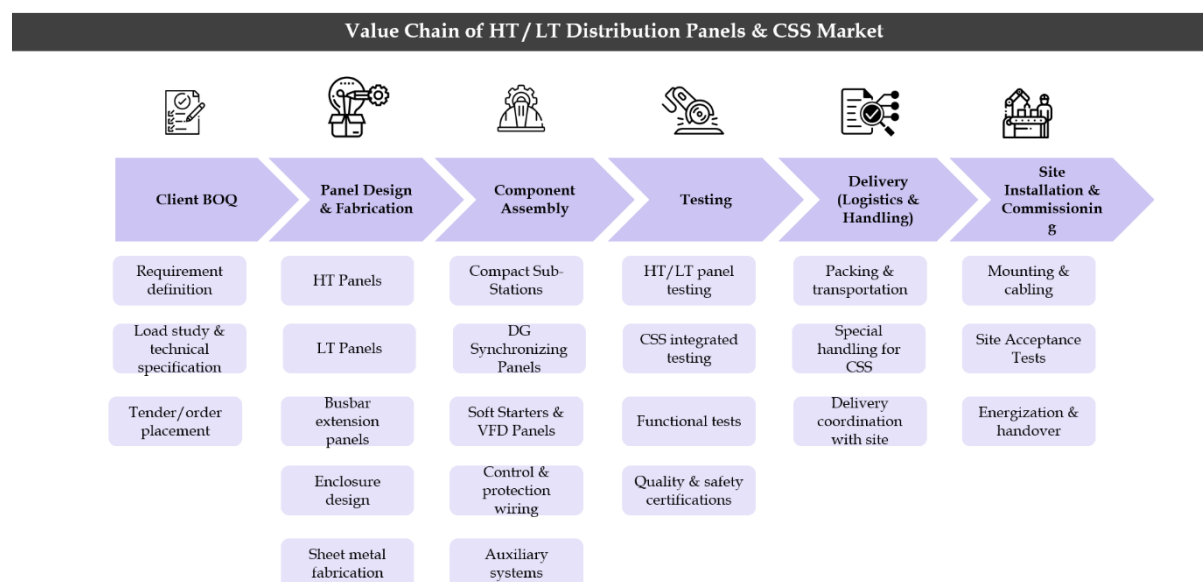


Source: Ken Research Analysis

3.4. VALUE CHAIN OF INDIA'S POWER DISTRIBUTION PANELS AND COMPACT SUBSTATION MARKET

The value chain of **Installing High Tension (HT) / Low Tension (LT) Distribution Panels and Compact Substation (CSS)** in India:

- Begins with the Client Bill of Quantities (BOQ, which refers to document used in construction and infrastructure projects that lists out the materials, parts, labor, and associated costs required to execute the work), which **outlines requirement** definition, technical specifications, and tender placement.
- Once the contract is awarded, the process moves to Panel Design & Fabrication and Component Assembly, covering HT/LT panels, Compact Substations, Diesel Generator (DG) synchronizing panels, and Variable Frequency Drive (VFD) panels.
- This is followed by Testing with functional validation and certifications, Delivery with logistics coordination, and
- Final stage Site Installation & Commissioning through cabling, energization, and handover.

Figure 3-4: Value Chain of India's Power Distribution Panels & Compact Substation Market

Source: Ken Research Analysis

Note: BOQ refers to Bill of Quantities, HT Panels refers to High Tension Panels, LT panels refer to Low Tension Panels, DG refers to Diesel Generator, VED refers to Variable Frequency Drive, CSS refers to Compact Substation.

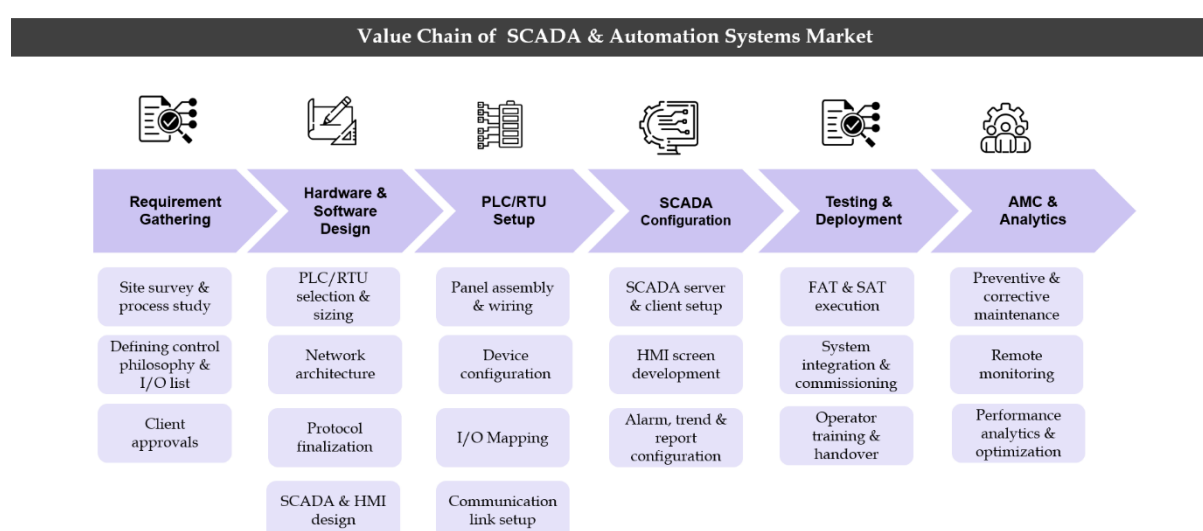
3.5. VALUE CHAIN OF INDIA'S SCADA & AUTOMATION SYSTEMS MARKET

The value chain of **SCADA and Automation Systems projects in India** primarily used in water treatment and power plants typically includes the following process flow:

- Requirement gathering through site surveys, process studies, and client approvals **to define scope and automation needs.**
- This is followed by hardware and software design, involving PLC/RTU selection, network architecture, and SCADA/HMI dashboards tailored to water quality, flow, and pumps in WTPs or turbines, boilers, and load management in power plants.

- Next comes PLC/RTU setup with panel assembly, wiring, instrumentation, and configuration of process logic such as dosing, backwashing, or turbine control.
- SCADA configuration then establishes servers, databases, alarms, and operator HMIs for centralized monitoring and control.
- The system is validated through Factory and Site Acceptance Tests (FAT & SAT) before moving into annual maintenance contracts and analytics, which enable preventive maintenance, remote monitoring, and performance optimization to ensure efficiency, reliability, and compliance.

Figure 3-5: Value Chain of India's SCADA & Automation Systems Market



Source: Ken Research Analysis

Note: SCADA refers to Supervisory Control and Data Acquisition, PLC refers to Programmable Logic Controller, RTU refers to Remote Terminal Unit, FAT refers to Factory Acceptance Test, SAT refers to Site Acceptance Test, AMC refers to Annual Maintenance Contract and I/O mapping refers to Input/Output Mapping, HMI refers to Human Machine Interface

4. INDIA'S RAILWAY EPC MARKET ANALYSIS

The Railway EPC Market is defined as the domestic revenue generated from EPC contracting activities related to railway electrification, signaling & telecommunication, and civil works in India. This includes electrification works such as OHE, TSS, and PSI; signaling & telecommunication systems; and civil works including station redevelopment, bridges, and platforms, along with associated O&M services where executed as part of EPC contracts.

4.1. OVERVIEW OF INDIA'S RAILWAY EPC MARKET

India's Railways are largely shaped by government budgets, with the capital outlay crossing **INR 2.6 lakh crore in FY26 Budget Estimates, driving most investments**. This sustained funding ensures a robust pipeline of projects in **electrification, signaling, and core civil infrastructure works** such as stations, depots, and bridges, defining the overall pace and scale of EPC opportunities.

The **Railway EPC market** covers projects grouped into **3 categories** (excluding track works and locomotive manufacturing works):

- **Civil Engineering Works:** Involves building core railway infrastructure through construction of stations, yards and bridges, along with supporting structures such as platforms, depots, warehouses, drainage and boundary walls.
- **Electrical Engineering Works (incl. Transmission and Distribution):** Provides traction and auxiliary power by undertaking electrification through 25 kV AC OHE,

traction substations, SCADA-enabled protection, and distribution networks across stations and depots.

- **Signal and Telecommunication Works:** Ensures safe and efficient train operations by deploying electronic/relay interlocking, block signalling, track circuit interfaces, and fibre/copper-based data networks.

These works are **fundamental to enhancing operational efficiency, safety and service reliability** in the Indian Railways, enabling seamless passenger and freight movement.

4.2. INDIA' RAILWAY EPC MARKET SIZE ANALYSIS, FY20–FY30F

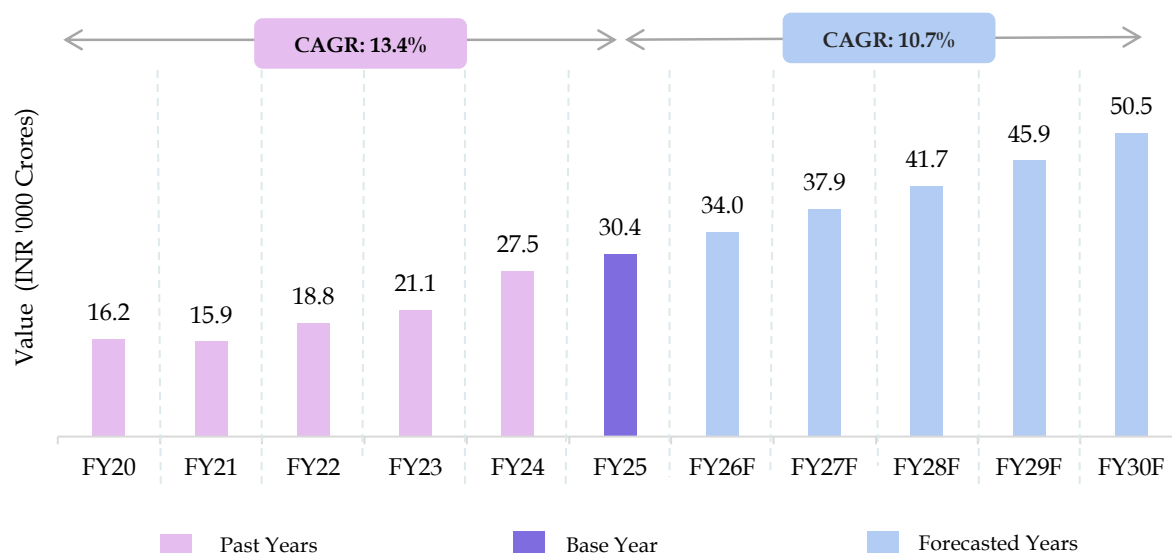
Between **FY20 and FY25**, the Railway EPC market increased from **INR 16.2 thousand crore to INR 30.4 thousand crore**, at a **CAGR of 13.4%**. Growth was driven by **increasing capital outlay under the National Rail Plan, the PM Gati Shakti programme**, and major projects such as the **Dedicated Freight Corridors**. Significant progress was also achieved in **station redevelopment, depots, bridges, and electrification with SCADA-enabled substations**.

The Speed Improvement Scheme and the freight capacity push ~3,000 MT together expand Indian Railways' EPC pipeline—lifting speeds to 120–160 km/h which are driving major EPC demand through **large-scale track, signaling, and OHE upgrades**, along with new freight terminals and logistics hubs—enhancing speed, capacity, and network efficiency across Indian Railways.

From **FY25 to FY30F**, the market is expected to grow from **INR 30.4 thousand crore to INR 50.5 thousand crore**, at a

CAGR of 10.7%. Key drivers include conclusion of **high-speed rail projects and their commissioning**, **electrification of remaining lines to meet the 100% electrification goal**, **OHE and substation upgrades**, **station modernization**, and further execution of **signaling programmes such as *Kavach***. Integration with **multimodal logistics hubs under Gati Shakti** will further support EPC demand.

Figure 4-1: India's Railway EPC Market Size in Terms of Value (in INR '000 Crore), FY20-FY30F



Source: Indian Railways Annual Reports, Ken Research Analysis

Note: 1: FY represents Financial Year (April-March)

Note 2: F represents Forecasted figures

The moderation in CAGR reflects a shift from policy driven budgetary expansion in FY20 to FY25 to execution focused growth in FY25 to FY30F, ensuring steady opportunities across civil, electrical and signaling works. As the demand for such projects increases, a direct increase for **railway electrification packages, overhead equipment, traction substation systems, signaling and telecom works and station redevelopment works** can be expected in the market.

4.3. INDIA'S RAILWAY EPC MARKET SEGMENTATION, FY25-FY30F

India's Railway EPC Segmentation by Work Type, FY25-FY30F:

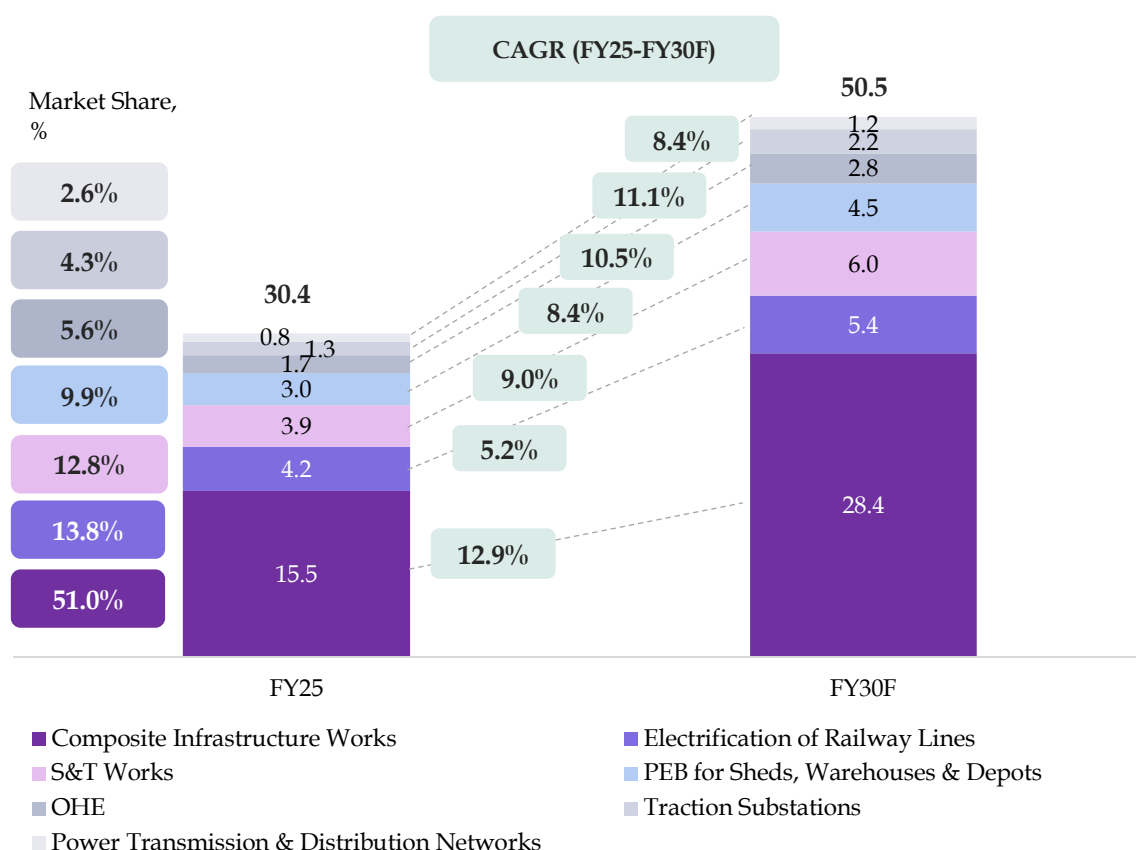
The Railway EPC market is concentrated in **composite infrastructure works**, which accounted for **51.0% of the market in FY25 at INR 15.5 thousand crore**. The growth in this segment is propelled by **new line additions, rising track density, and stronger safety and security requirements**, alongside investments in **station redevelopment and passenger amenities**. The segment is expected to lead, reaching **INR 28.4 thousand crore by FY30F**, driven by modernization programs, urban station upgrades, and capacity expansion.

The second-largest segment is **electrification of railway lines**, contributing **13.8% of the market in FY25 at INR 4.2 thousand crore**. Growth in this segment showing signs of saturation, due to accelerated electrification programs undertaken between FY20 and FY25 to reduce fuel costs and promote sustainability. **As the network nears complete electrification, growth is expected to slow**, with the segment projected to reach **INR 5.4 thousand crore by FY30F at a CAGR of 5.2%**.

Signaling & Telecommunication (S&T) works accounted for 12.8% of the market in FY25 at INR 3.9 thousand crore. Growth is being driven by safety programs such as Kavach and automatic block signaling, improving efficiency and reducing collisions. The segment is projected to reach INR 6.0 thousand crore by FY30F at an 9.0% CAGR, supported by wider adoption of digital signaling and upgrades across busy corridors.

Among the faster-growing but smaller segments, **traction substations** (CAGR 11.1%) and **OHE works** (CAGR 10.5%) are expected to see strong growth through FY30F, driven by the ongoing need to expand and modernize traction systems and feeder lines to support higher-capacity, fully electrified corridors.

Figure 4-2: India's Railway EPC Market Segmentation (INR '000 crore) by Work Type, FY25 & FY30F



Source: Ken Research Analysis

Note: F represents Forecasted figures

4.4. OPPORTUNITIES IN INDIA'S RAILWAY EPC MARKET

India's Railway EPC market was valued at **INR 30.4 thousand crore in FY25** and is projected to reach **INR 50.5**

thousand crore by FY30F, growing at a CAGR of 10.7%. Opportunities for EPC players lie in the **execution of electrification, signaling, and station redevelopment works** under flagship programs such as **PM Gati Shakti, Amrit Bharat Station Scheme, and Mission Electrification.** These initiatives are driving large-scale modernization of railway infrastructure, including depots, bridges, and traction substations, to enhance operational efficiency and connectivity across corridors.

In addition, **high-speed rail corridors, dedicated freight lines, and multimodal logistics hubs** are creating sustained EPC demand across civil and electrical packages. The push toward **100% rail electrification and digital signaling integration** is expanding opportunities in **OHE systems, SCADA-enabled substations, and depot construction.** As Indian Railways continues to prioritize **capacity expansion and asset modernization,** EPC players with capabilities in **integrated design, automation, and turnkey execution** are well-positioned to capture value across this growing market.

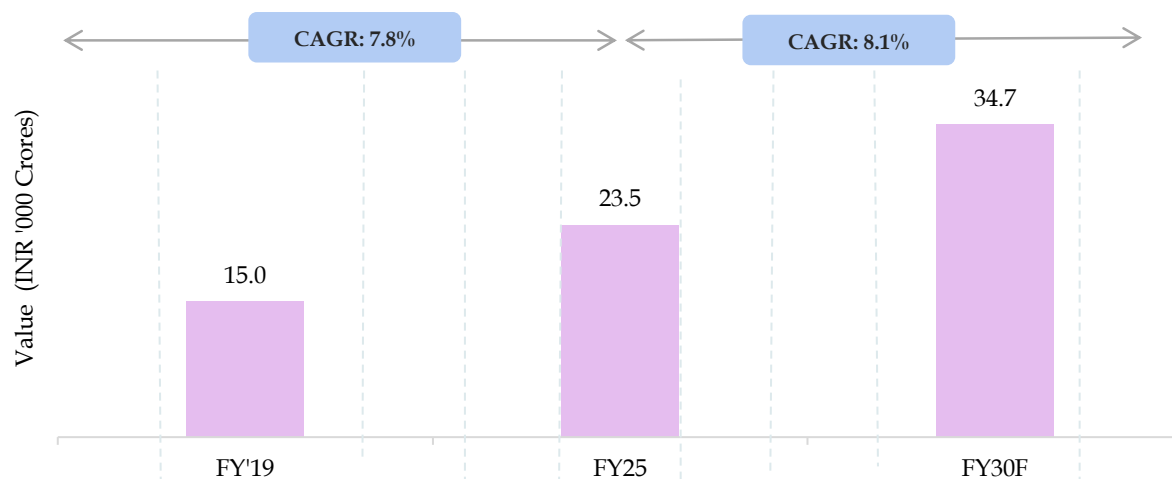
4.5. INDIA METRO SYSTEMS EPC MARKET SIZE ANALYSIS, FY19-FY25-FY30F

The **India Metro Systems EPC Market** is defined as the domestic revenue generated from EPC contracting activities related exclusively to urban metro rail systems in India. This includes EPC execution for metro-specific civil infrastructure (elevated and underground corridors, viaducts, stations, depots, tunnels), metro rolling-stock-independent systems such as traction power supply, metro signaling & telecommunication systems (including CBTC), platform screen doors, depot systems, and associated electromechanical works, along with O&M services where bundled within metro EPC contracts.

India's metro EPC market has grown steadily from **₹15,000 crore in FY19** to an estimated **₹23,500 crore in FY25**, reflecting a CAGR of 7.8%, and is projected to reach **₹34,700 crore by FY30F**, reflecting a **8.1% CAGR**. This moderate yet sustained expansion highlights consistent investments in urban transport infrastructure, driven by new corridors and extensions across major cities such as Delhi, Mumbai, Bengaluru, Pune, and Ahmedabad.

According to the **Ministry of Housing and Urban Affairs (MoHUA)**, over **990 km** of metro projects have been approved since 2014, while an additional **1,000 km** is under various stages of planning and construction across **27+ cities**. The upcoming pipeline, valued at approximately **₹3 lakh crore**, supports a robust EPC opportunity averaging **₹25,000–35,000 crore annually** through FY30F.

This growth reflects a balanced shift toward more complex underground alignments and integrated transit systems. However, execution challenges – land acquisition, funding delays, and rising input costs – may constrain completion rates. Overall, India's metro EPC market will remain a key infrastructure driver under the **Vision 2047 Urban Mobility** agenda, underpinned by steady public spending and state-level project rollouts.

Figure 4-3: India's Metro System EPC Market Size in Terms of Value (in INR '000 Crore), FY19-FY25-FY30F

Source: Ministry of Housing and Urban Affairs, Ken Research Analysis

Note 1: FY represents Financial Year (April-March)

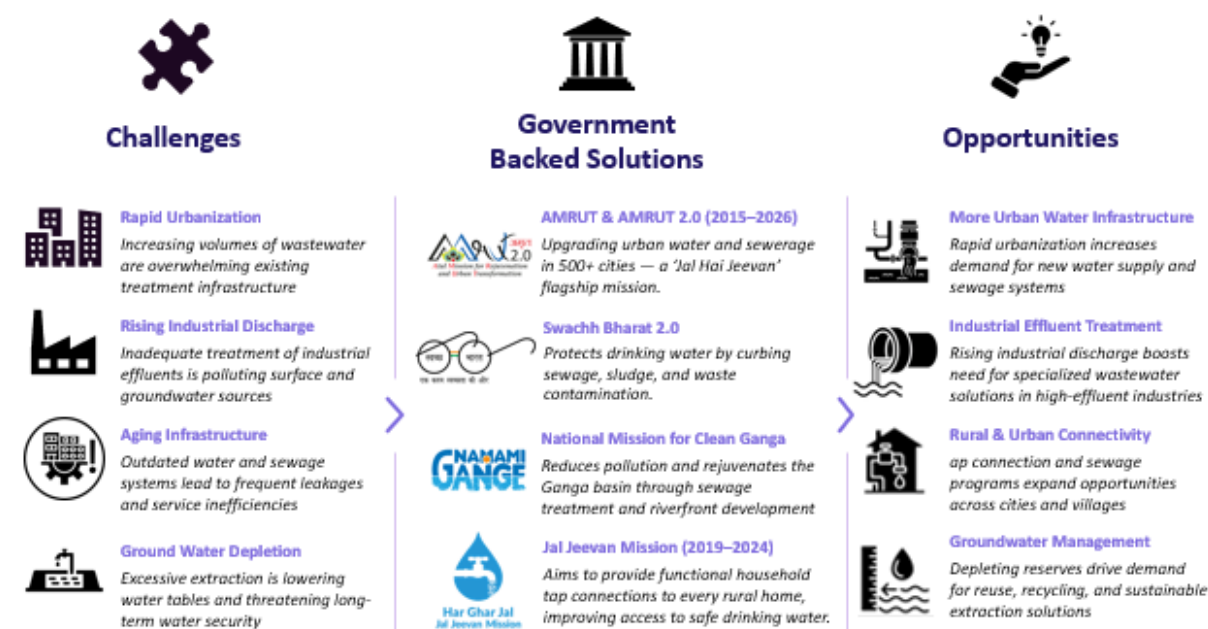
Note 2: F represents Forecasted figures

5. INDIA'S DRINKING WATER AND WASTE WATER TREATMENT MARKET ANALYSIS

The Drinking Water and Waste Water Treatment Market is defined as the domestic EPC revenue generated from the design, supply, and execution of drinking water treatment plants, wastewater/sewage treatment plants, distribution networks, pumping stations, and electro-mechanical works in India. Distribution networks include pipelines, OHTs, and reservoirs.

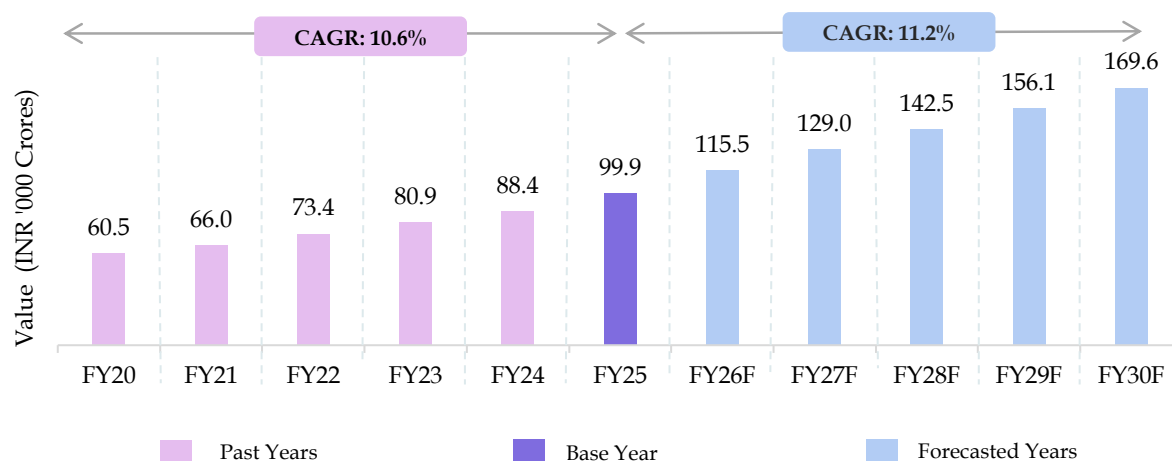
5.1. INDIA'S DRINKING WATER AND WASTE WATER TREATMENT MARKET, FY20-FY30F

This market covers EPC services for water supply, sewage, and effluent treatment plants and related civil and electromechanical works in India. India continues to face critical challenges in drinking water access and wastewater management. To address these challenges, the government has rolled out a series of large-scale programs focused on expanding piped water access, modernizing urban infrastructure, and promoting wastewater treatment and reuse.

Figure 5-1: Challenges and Opportunities in India's Drinking Water and Waste Water Market

Source: Ken Research Analysis

These efforts have driven strong market growth, with India's drinking water and wastewater treatment investments expanding from **INR 60.5 thousand crore in FY20** to **INR 99.9 thousand crore in FY25**, at a **CAGR of 10.6%**. The momentum reflects increasing demand for EPC execution and professional O&M services across both urban and industrial segments.

Figure 5-2: India's Drinking Water and Waste Water Treatment Market in Terms of Value (in INR '000 Crore), FY20-FY30F

Source: Ken Research Analysis

Note: 1: FY represents Financial Year (April-March)

Note 2: F represents Forecasted figures

Looking ahead, the sector is expected to continue its strong growth as India's wastewater challenges remain unresolved, reaching **INR 169.6 thousand crore by FY30F at a CAGR of ~11.2%**, supported by:

- Continued rollout of **Jal Jeevan Mission** and **AMRUT 2.0** in underserved regions
- Expansion of **wastewater reuse policies** under the **National Mission for Clean Ganga** and **National Water Policy**
- Expansion of **Swachh Bharat Mission 2.0** and CMS-led sanitation programs, driving advanced waste and wastewater management systems.
- Stricter industrial compliance enforced by **CPCB** and **SPCBs**
- Rising adoption of **advanced treatment technologies** and **digital O&M platforms**
- Increasing private sector participation in **design-build-operate** contracts

EPC and O&M Market of India's Drinking Water and Waste Water Treatment Market

India's drinking water and wastewater treatment market is driven by two spending streams: **capital expenditure (CAPEX)** for setting up plants and infrastructure, and **operational expenditure (OPEX)** for their upkeep.

- CAPEX includes core plant construction along with intake systems, pump houses, reservoirs, and pipelines,
- OPEX covers recurring costs of operation, maintenance, and compliance.

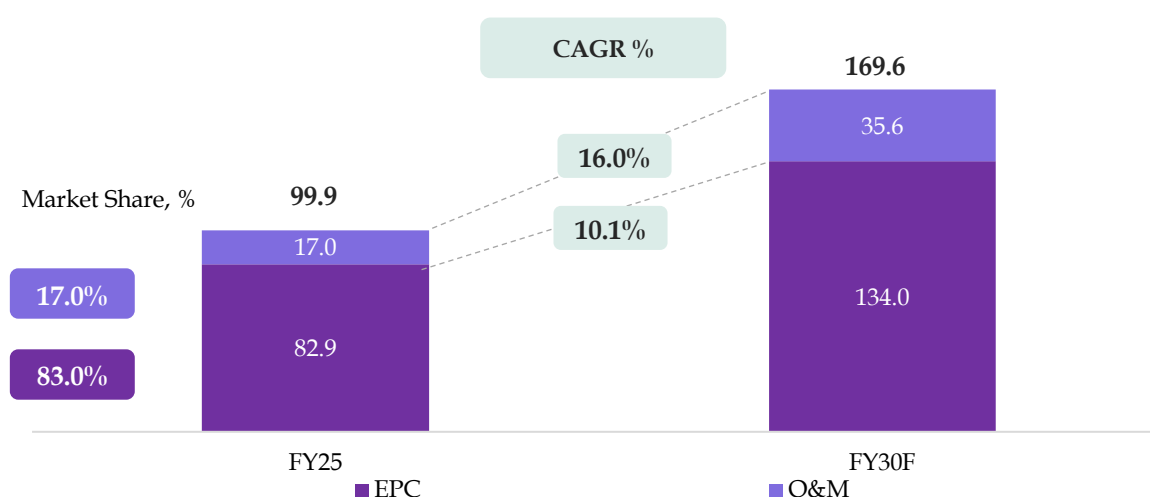
These translate into two commercial segments: **Engineering, Procurement & Construction (EPC)**, reflecting CAPEX-led demand for new assets, and **Operations & Maintenance (O&M)**, reflecting OPEX-led lifecycle services. Together, they provide a full view of the market, balancing cyclical construction activity with stable O&M demand.

As more opportunities open up for EPC and O&M players in the market, there exists a variation in the opportunities available. From FY20 to FY25, India's drinking water and wastewater treatment market was **driven by EPC spend, while O&M formed a smaller share.** In FY20, EPC contributed 80% of the market, with O&M at 20%. **This trend continued through FY25, with EPC rising to 83%,** supported by large-scale project rollouts under Jal Jeevan Mission and AMRUT 2.0 growing at a **CAGR of 10.1%** from FY25-FY30F.

As these schemes are scheduled to conclude by FY26, a shift is expected post-implementation, with increased O&M spend to operate and maintain commissioned assets. By FY30F, O&M is projected to rise to 21%,

reflecting growing demand for professional long-term operations reflecting a **CAGR of 16.0%** from FY25-FY30F.

Figure 5-3: India's Drinking Water and Waste Water Treatment Market Share Split between EPC and O&M (INR '000 crore), FY25 - FY30F



Source: Ken Research Analysis

Note: F represents Forecasted figures,

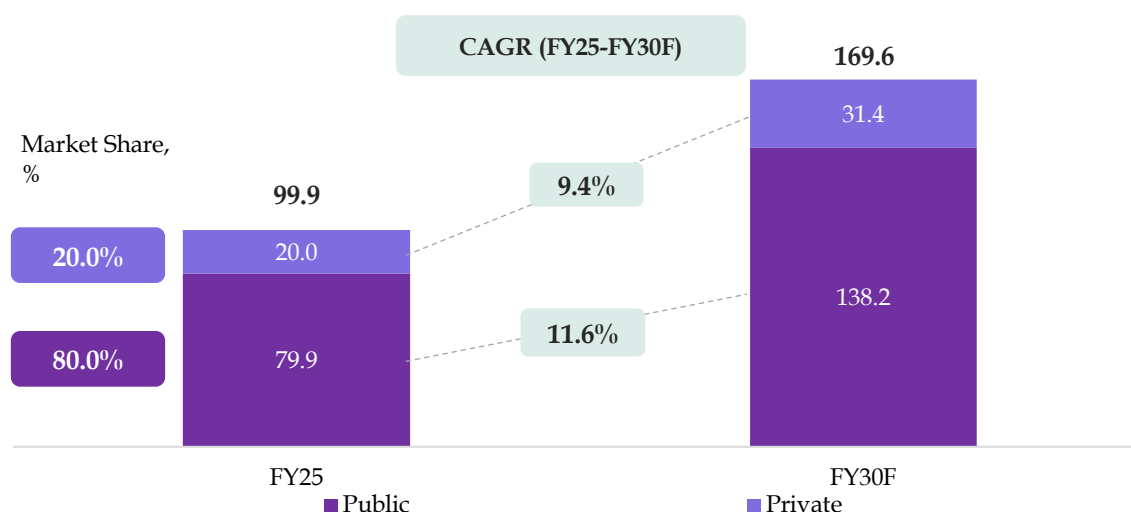
5.2. INDIA'S DRINKING WATER AND WASTE WATER TREATMENT MARKET SEGMENTATION, FY25-FY30F

India's Drinking Water and Waste Water Treatment Market Segmentation by End User, FY25-FY30F:

The Indian water and wastewater infrastructure Market is driven by the public domain, accounting for 80% of the market in FY25 at INR 79.9 thousand crore. This segment majorly includes municipal water infrastructure including drinking water treatment plants, sewage treatment plants, sewage and water piping and more. It is largely funded through government programs like Jal Jeevan Mission, AMRUT, and Smart Cities, and is projected to grow at a 11.6% CAGR to INR 138.2 thousand

crore by FY30F, supported by rural water access, sewage expansion, and urban sanitation upgrades.

Figure 5-4: India's Drinking Water and Waste Water Treatment Market Segmentation (INR '000 crore) by End User, FY25 & FY30F



Source: Ken Research Analysis

Note: F represents Forecasted figures

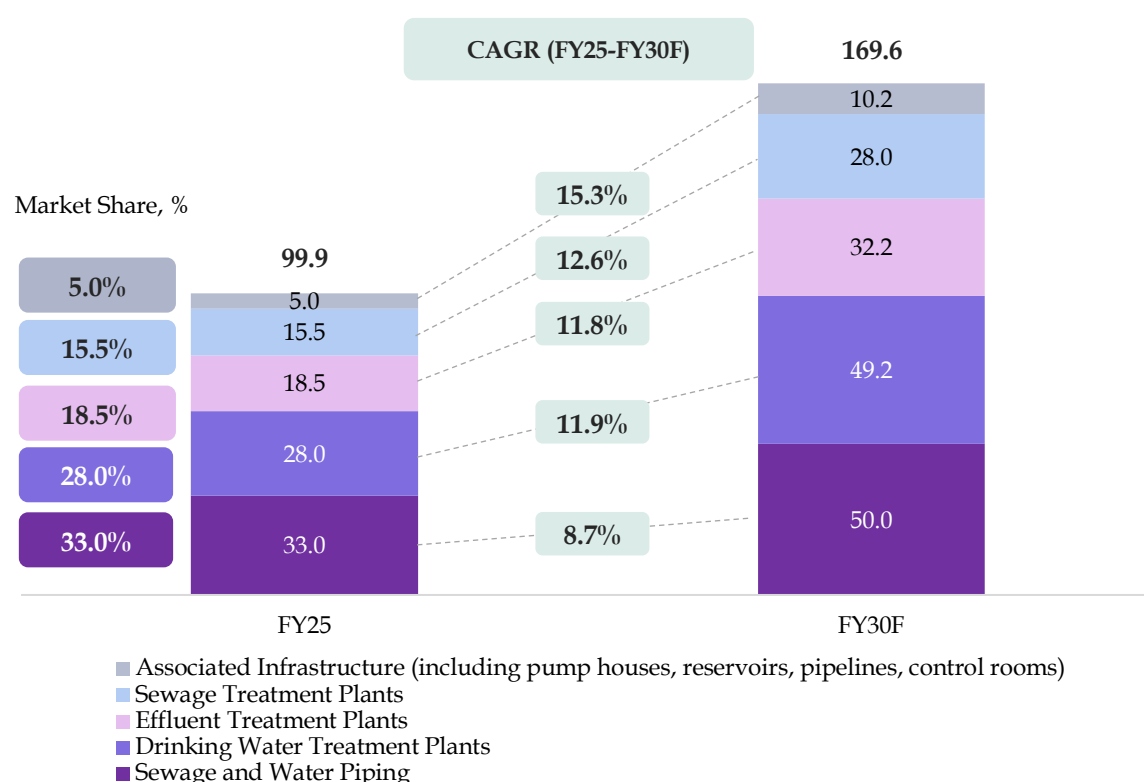
The private and industrial domain comprised the remaining 20% at INR 20 thousand crore, majorly covering industrial water infrastructure including effluent treatment plants (ETPs), select associated infrastructure, and private sewage and water piping distribution projects. Growth in this segment is driven by regulatory compliance, industrial expansion, and adoption of reuse technologies. It is expected to expand at an 9.4% CAGR to INR 31.4 thousand crore by FY30F

India's Drinking Water And Waste Water Treatment Market Segmentation by Plant Type, FY25-FY30F:

The largest segment is **Sewage and Water Piping**, which contributed 33.0% of the market in FY25 at INR 33.0 thousand crore. Although its share is expected to moderate by FY30F, the segment will still expand to INR 50.0

thousand crore, growing at a CAGR of 8.7%. Demand is sustained by ongoing expansion of water distribution and sewerage networks, even as treatment capacity becomes a greater focus.

Figure 5-5: India's Drinking Water and Waste Water Treatment Market Segmentation (INR '000 crore) by Plant Type, FY25 & FY30F



Source: Ken Research Analysis

Note: F represents Forecasted figures, Associated Infrastructure refers to the Balance of Plant which is required for construction and includes pump houses, reservoirs, control rooms etc.

Drinking Water Treatment Plants (DWTPs) formed 28.0% of the market in FY25 and is forecasted to reach INR 49.2 thousand crore by FY30F, growing at 11.9% CAGR. Government-led rural supply programs such as Jal Jeevan Mission and AMRUT 2.0, alongside rising urban and industrial water needs, are the main drivers. Similarly, **Effluent Treatment Plants (ETPs)** accounted for 18.5% in FY25 (INR 18.5 thousand crore) and are projected to

expand at 11.8% CAGR to INR 32.2 thousand crore by FY30F, supported by stricter regulations, Zero Liquid Discharge (ZLD) adoption, and rising industrial wastewater volumes.

Though smaller in scale, **Associated Infrastructure**, which includes pump houses, reservoirs, pipelines, and control rooms is set to grow the fastest. Valued at INR 5.0 thousand crore in FY25, it is expected to nearly double to INR 10.2 thousand crore by FY30F at 15.3% CAGR. This reflects the sector's emphasis on integrated systems, reliability, and lifecycle efficiency of water assets.

5.3. OPPORTUNITIES IN INDIA'S WATER & WASTEWATER TREATMENT MARKET

The Indian Drinking Water and Wastewater Treatment market, valued at **INR 99.9 thousand crore in FY25**, is projected to reach **INR 169.6 thousand crore by FY30F**, growing at a **CAGR of ~11.2%**. Opportunities for EPC players lie in the design and execution of **drinking water, sewage, and effluent treatment plants** under flagship government programs such as **Jal Jeevan Mission, AMRUT 2.0, and Namami Gange**. The expansion of municipal and industrial infrastructure, supported by stricter environmental regulations, continues to drive EPC demand across urban and rural India.

Rising **urbanization, industrial compliance norms, and Zero Liquid Discharge (ZLD)** requirements are accelerating investments in **STPs, ETPs, and reuse systems**, while **smart water infrastructure and SCADA-enabled O&M contracts** provide recurring opportunities for **technology-driven EPC and system integration firms**. With over **80% of projects led by public sector clients**, the sector offers sustained visibility and long-term

engagement potential across both **capital execution and lifecycle management** services.

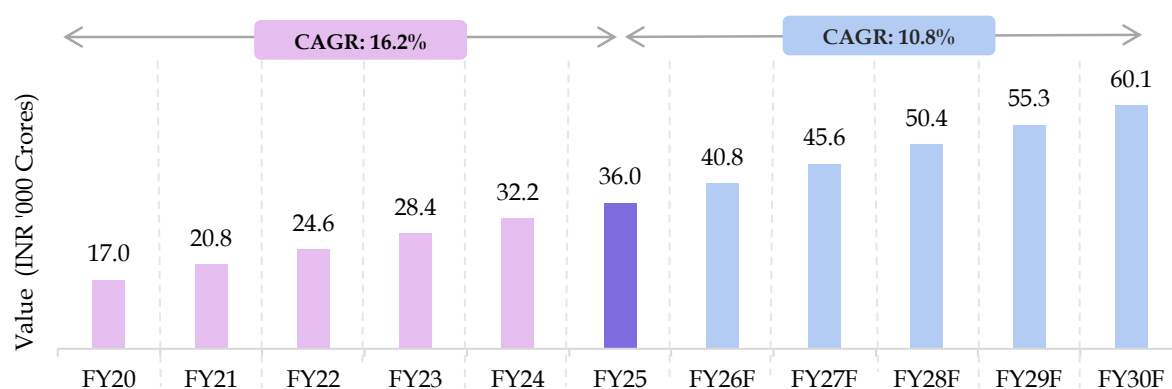
6. INDIA'S TRANSMISSION AND DISTRIBUTION MARKET ANALYSIS

The **Transmission and Distribution (T&D) Market** is defined as the domestic EPC revenue generated from the design, supply, construction, and commissioning of transmission lines, substations, and distribution networks across India. The scope includes high-voltage (HV) and medium-voltage (MV) infrastructure supporting power evacuation, grid interconnection, and last-mile distribution.

6.1. OVERVIEW OF INDIA'S TRANSMISSION AND DISTRIBUTION MARKET AND ITS MARKET SIZE, FY20-FY30F

India's power sector is transforming rapidly with rising electricity demand and large-scale renewable integration. To strengthen grid reliability, the government has launched major initiatives to expand interregional **transmission corridors**, **modernize distribution networks**, and promote smart metering and loss reduction. These efforts, aligned with the target of integrating 500 GW of renewables by 2030, are driving sustained investment across the transmission and distribution sector.

Figure 6-1: India's Transmission and Distribution Market in Terms of Value (in INR '000 Crore), FY20-FY30F



 Past Years Base Year Forecasted Years

Source: Ken Research Analysis

Note: 1: FY represents Financial Year (April-March)

Note 2: F represents Forecasted figures

As of **FY25**, India's **transmission and distribution market size** stood at **INR 36,000.0 crore**, growing at a strong **CAGR of 16.2 %** over FY20–FY25. Growth was fuelled by accelerated grid expansion, post-COVID catch-up investments, and government initiatives to strengthen corridors, modernize distribution, deploy smart metering, and reduce losses, along with increased private participation in transmission.

The market is projected to reach **INR 60,063.0 crore** by **FY30F**, at a **CAGR of 10.8 %**, reflecting somewhat moderated growth. Although the government has laid out an **INR 9 lakh crore transmission plan (2023–2032)**, past under-execution of targets is expected to persist. Still, sustained investment will continue, driven by ambitions to integrate 500 GW of renewables, modernize grid assets, and enhance reliability. The National Electricity Plan envisages addition of **1.91 lakh circuit kilometres** of transmission lines by 2032 over the period from 2023 to 2032, expanding the network to ~6.48 lakh ckm.

6.2. INDIA'S TRANSMISSION AND DISTRIBUTION MARKET SEGMENTATION, FY25-FY30F

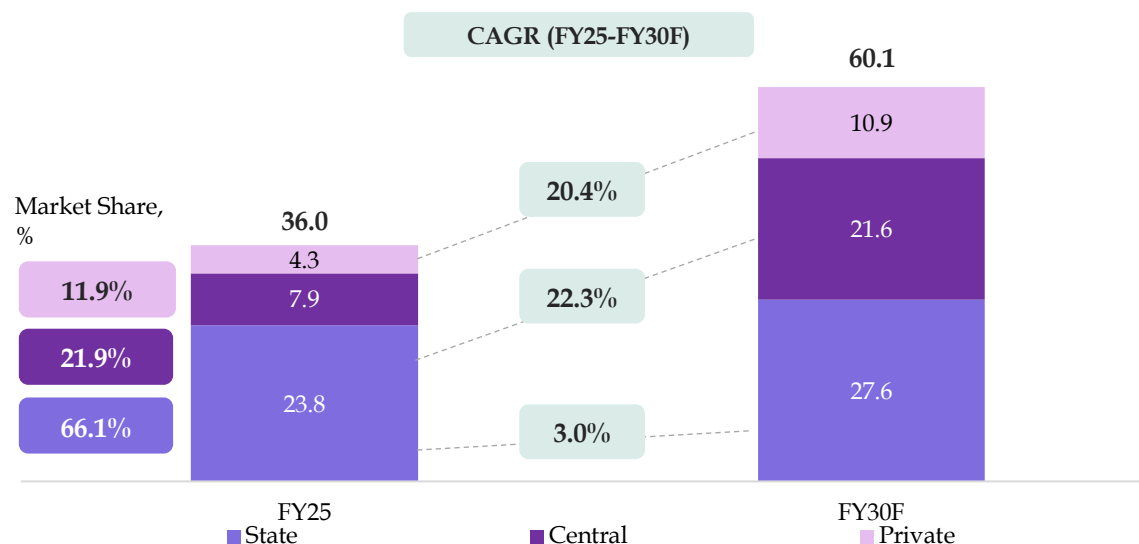
India's Transmission and Distribution Market Segmentation by End User, FY25-FY30F:

The **State segment** was the largest in **FY25**, with a market size of **INR 23.8 thousand crore**, driven by state utility and municipal investments in grid modernization, feeder segregation, smart metering, and **urban infrastructure** such as **water and wastewater treatment facilities**.

Growth is expected to **moderate**, with a **CAGR of 3.0%**, reaching **INR 27.6 thousand crore** by **FY30F**, as fiscal constraints and the shift of large projects towards central and private players halt state expansion.

The **Central segment** stood at **INR 7.9 thousand crore** in **FY25** and is projected to grow rapidly at a **CAGR of 22.3%** to **INR 21.6 thousand crore** by **FY30F**. Growth is expected to **accelerate** due to central-led renewable evacuation and interregional transmission projects under the **INR 9 lakh crore national plan**, supported by faster approvals and standardized bidding.

The **Private segment**, including industrial and independent developers, was **INR 4.3 thousand crore** in **FY25** and is expected to grow at a **CAGR of 20.4%** to **INR 10.9 thousand crore** by **FY30F**. The segment is expected to **emerge as one of the most lucrative**, driven by growing participation in tariff-based competitive bidding, rising industrial demand, and investments in grid digitalization and automation.

Figure 6-1: India's Transmission and Distribution Market Segmentation (INR '000 crore) by End User, FY25 & FY30F

Source: Ken Research Analysis

Note: F represents Forecasted figures

6.3. OPPORTUNITIES IN INDIA'S TRANSMISSION AND DISTRIBUTION MARKET

India's Transmission and Distribution market, valued at **INR 36.0 thousand crore in FY25**, is projected to reach **INR 60.1 thousand crore by FY30F**, reflecting a **CAGR of 10.8%**. Opportunities for EPC players lie in **grid expansion, substation construction, and feeder modernization** driven by government programs such as the **Revamped Distribution Sector Scheme (RDSS), Green Energy Corridor, and National Electricity Plan**. The large-scale push for **renewable evacuation and interregional transmission corridors** continues to anchor steady EPC demand across both state and central utilities.

Further opportunities emerge from the **growing need for reliable power transmission** to support **water, industrial, and renewable infrastructure**, encouraging development

of **integrated T&D solutions**. Rising **private sector participation through tariff-based competitive bidding (TBCB)** and **O&M contracts** is expanding the addressable market for **agile EPC firms** specializing in **electrical, automation, and grid modernization** projects. This diversification—across state, central, and private segments—positions capable EPC players to capture sustained, high-value contracts through FY30F.

7. INDIA'S POWER DISTRIBUTION PANELS AND COMPACT SUB- STATIONS (CSS) MARKET ANALYSIS

The Power Distribution Panels and Compact Substation Market is defined as the domestic value generated from the assembly and installation of HT/LT distribution panels and compact substations (CSS) within India. This includes panels and substations deployed as part of EPC projects in Railways, Water, and other industries, as well as standalone supply to other contractors.

7.1. OVERVIEW OF INDIA'S POWER DISTRIBUTION PANELS AND COMPACT SUBSTATIONS MARKET AND ITS MARKET SIZE, FY20-FY30F

India's Power Distribution Panel and Compact Substation Market include differently rated panels: High Tension (HT, >33 kV) for grid-level distribution, Medium Tension (MT, 1-33 kV) for industrial and large commercial loads, and Low Tension (LT, <1 kV) for residential and small-scale use.

In FY20, the market was valued at **INR 105.1 thousand crore**, growing to **INR 144.5 thousand crore** by FY25 at a **CAGR of 6.6%**, driven by **electrification, utility investments, and industrial expansion**. By FY30F, it is projected to reach **INR 209.1 thousand crore**, growing at a **CAGR of 7.7%**, supported by **renewable energy parks, EV infrastructure, and Make-in-India manufacturing**.

Grid modernization is a key driver. The **National Electricity Plan (2023-2032)** targets expansion of transmission lines from **4.85 lakh ckm (circuit km)** to **6.48 lakh ckm**, and transformation capacity from **1,251 GVA** to **2,342 GVA**, including **33 GW of HVDC lines** and evacuation of **280 GW of renewable energy** (Source: Ministry of Power).

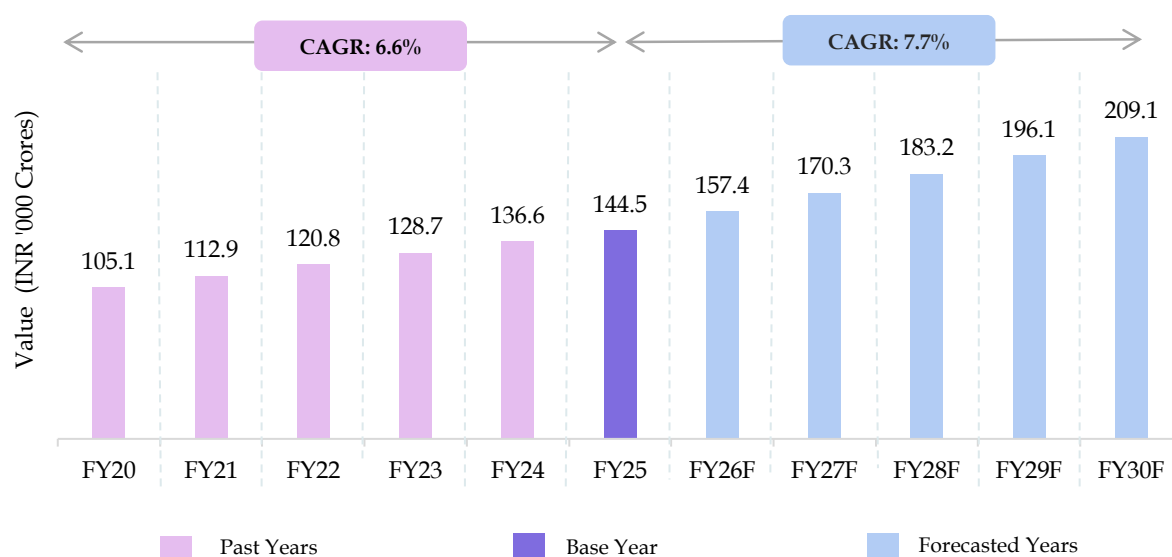
The **Ministry of Heavy Industries** has sanctioned over **9,000 public EV charging stations** under **FAME-II**, while broader estimates suggest India will require over **13**

lakh stations by 2030, boosting demand for LT and MT panels (Source: IBEF).

Further, Domestic manufacturing is gaining momentum, supported by **FDI inflows**. This Industrial sector growth in **cement, metals, and data centres** also continues to drive demand for **power distribution panels**.

Despite the persisting challenges in the market including **financial stress among DISCOMs, slower EV adoption, and transmission bottlenecks** affecting renewable energy evacuation. The market remains on a **strong growth trajectory**, backed by **structural reforms and policy support**.

Figure 7-1: India's Power Distribution Panels and Compact Substation Market Size in Terms of Value (in INR '000 Crore), FY20-FY30F



Source: Ken Research Analysis

Note 1: FY represents Financial Year (April-March)

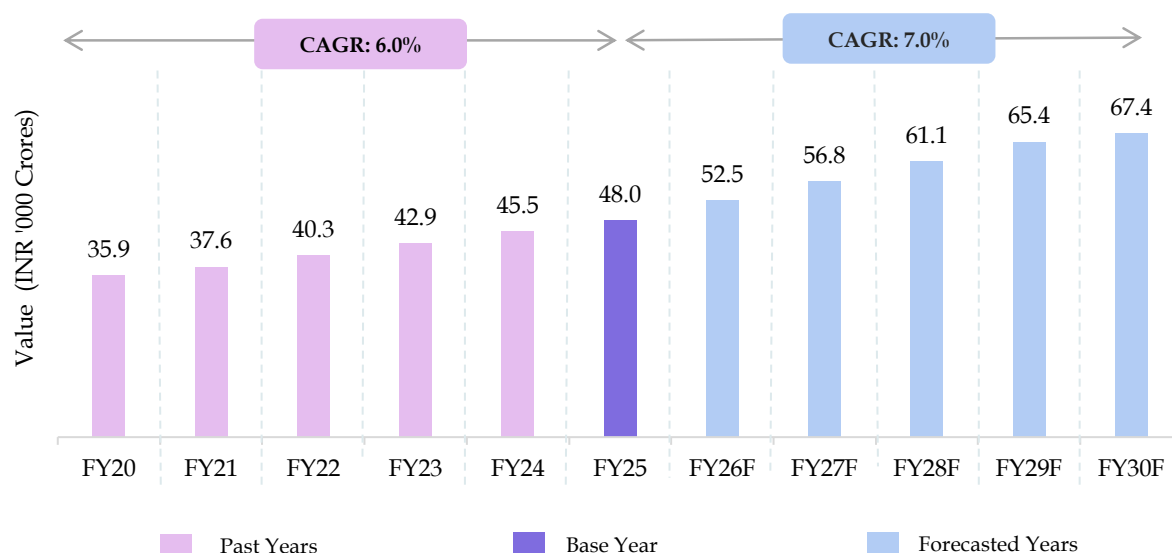
Note 2: F represents Forecasted figures

7.2. INDIA'S SWITCHGEAR INDUSTRY, FY20-FY30F

The India Switchgear Market represents the total revenue generated from the sale of electrical switchgear used to control, protect, and isolate power systems across transmission, distribution, and end-user installations. It products such as circuit breakers, panels, RMUs, and gas- or air-insulated switchgear supplied by domestic and international manufacturers. The market is measured as the value of switchgear products sold in India, expressed in **INR crores**.

Distribution panels represent the final assembly, within which **switchgear** is only one component alongside busbars, relays, meters, and enclosures. As a result, **the panel market is structurally larger and has consistently been at least 3X the size of the standalone switchgear market**, reflecting the added value of integration, customization, and ancillary equipment.

In FY20, the Indian switchgear market stood at **INR 35.9 thousand crore**, rising to **INR 48.0 thousand crore by FY25**, at a CAGR of **6.0%**. Growth during this period was steady, supported by demand from utilities and industrial projects, although moderated by pandemic-related slowdowns.

Figure 7-2: India's Switchgear Market Size in Terms of Value (in INR '000 Crore), FY20-FY30F

Source: Ken Research Analysis

Note: 1: FY represents Financial Year (April-March)

Note 2: F represents Forecasted figures

Looking ahead, the market is projected to reach **INR 67.4 thousand crore by FY30F**, growing at a CAGR of **7.0%** during FY25–FY30F. This stronger momentum will be supported by renewable energy integration, grid modernization, EV infrastructure, and industrial automation, similar to the drivers shaping growth in India's overall distribution panel market.

7.3. INDIA'S POWER DISTRIBUTION PANELS AND COMPACT SUBSTATION MARKET SEGMENTATION, FY25-FY30F

India's Power Distribution Panels and Compact Substations Market Segmentation by Type of Panel, FY25-FY30F:

India's Power Distribution Panels and Compact Substation (CSS) market is evolving with distinct growth patterns across segments, driven by infrastructure upgrades, industrial demand, and urban development.

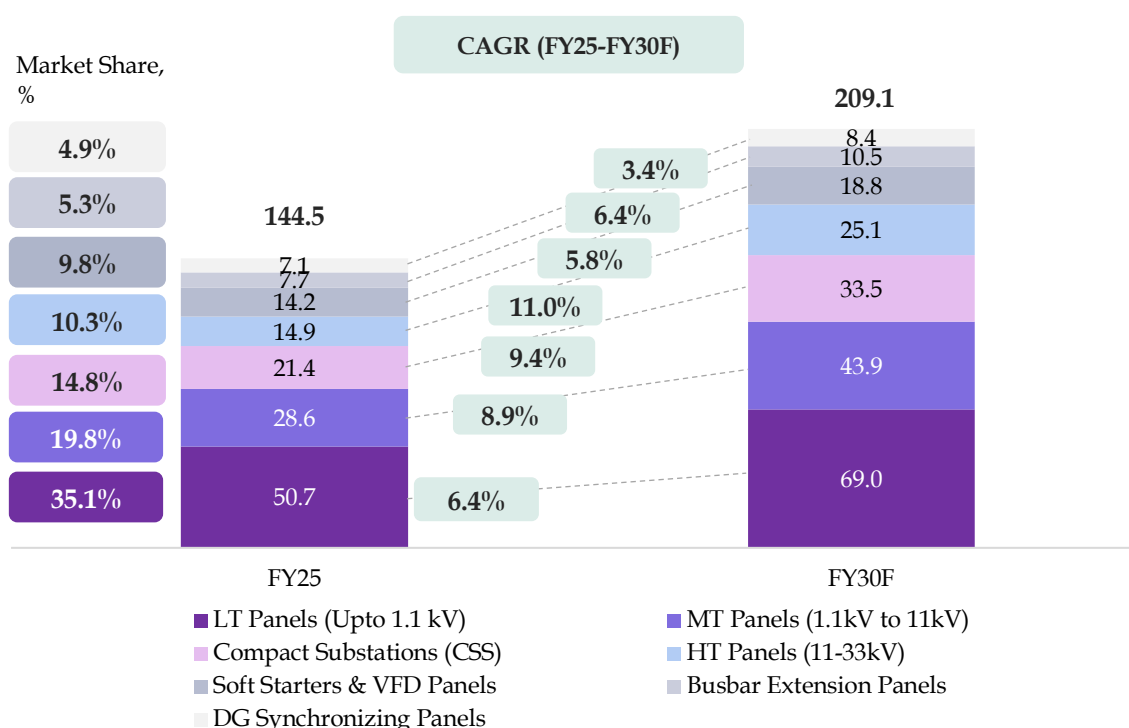
The **LT Panels (up to 1.1 kV)** segment remains the **largest**, contributing **35.1% in FY25 at INR 50.7 thousand crore** and is expected to grow at a CAGR of 6.4% between FY25 and FY30F, reaching a market size of **INR 69.0 thousand crore**. Growth is steady, supported by **residential, commercial, and real estate demand**, though the segment is maturing.

HT Panels (11-33 kV), while smaller in share at **10.3% in FY25 at INR 14.9 thousand crore**, is growing the fastest, with a CAGR of 11.0% by **FY30F, reaching INR 25.1 thousand crore**. Growth is led by **transmission upgrades, renewable evacuation, and state-level investments under RDSS**.

Compact Substations (CSS) contribute **14.8% in FY25 at INR 21.4 thousand crore** and is expected to reach **INR 33.5 thousand crore by FY30F** growing at a CAGR of 9.4% between FY25 and FY30F. This growth is supported by **smart cities, metro rail, and distributed solar infrastructure** in space-constrained environments.

Smaller segments like **Soft Starters & VFD Panels**, **DG Synchronizing Panels**, and **Busbar Extension Panels** show mixed trends. While **DG panels** are slowing due to grid reliability and emissions norms, **automation-focused panels** are growing with the shift toward **energy efficiency and intelligent distribution systems**.

Figure 7-3: India's Power Distribution Panels and Compact Substation Market Segmentation (INR '000 crore) by Panel Type, FY25 & FY30F



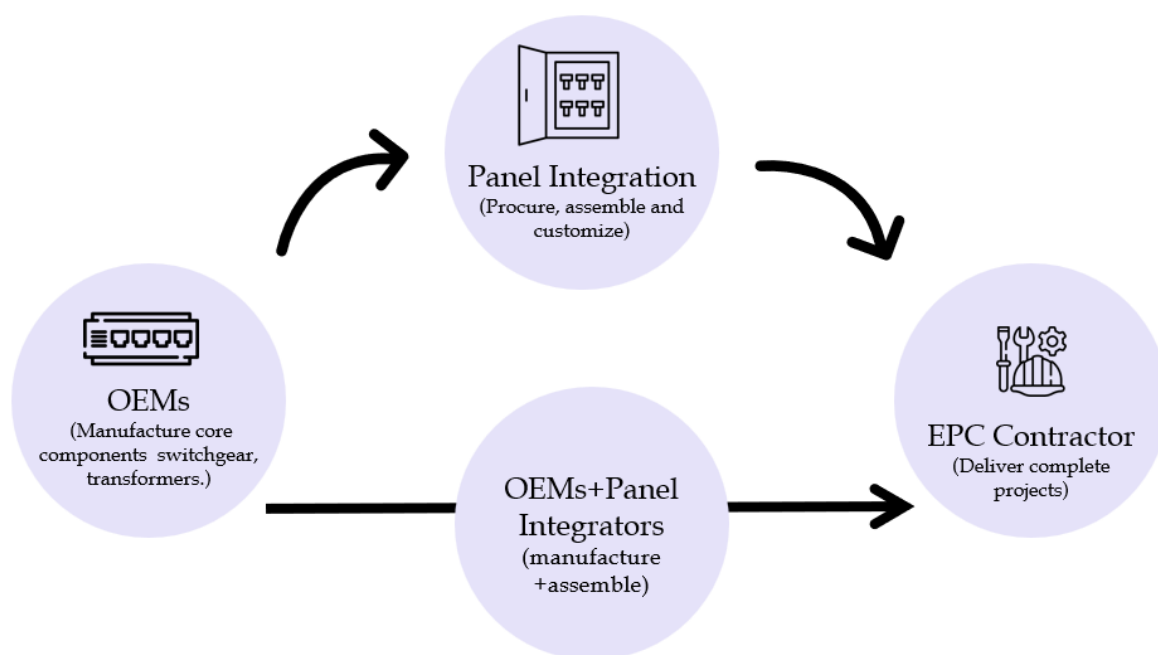
Source: Ken Research Analysis

Note: F represents Forecasted figures

7.4. MARKET STRUCTURE COMPRISES OF OEMs, PANEL INTEGRATION AND EPC CONTRACTORS

The market structure for HT/LT Distribution Panels and Compact Sub-Stations (CSS) is defined by the interaction between **OEMs** (Original Equipment Manufacturers), **Panel Integrators**, and **EPC (Engineering, Procurement & Construction) Contractors**. This interaction, can be seen below.

Figure 7-4: Interaction Between OEMs, Panel Integrators, and EPC Contractors



Source: Ken Research Analysis

- **OEMs** focus on manufacturing the core electrical components such as circuit breakers, relays, transformers, and switchgear.
- Now once components are manufactured there are 2 models seen in the market:
 - **OEMs + Panel Integrators:** The OEMs themselves use the manufactured

components to create power distribution panels to offer to the customers, and because of their larger footprint offer more expensive aftermarket services. And the other is,

- **Panel Integrators** players who procure components from OEMs and assemble the distribution panels as per customizations, and configurations required by customers. They compete against OEMs + Panel Integrators by offering less expensive panels
- **EPC Contractors** oversee the end-to-end project delivery, including design, engineering, procurement, installation, and commissioning, ensuring seamless integration of panels and substations within the overall electrical infrastructure.

This layered ecosystem ensures efficient manufacturing, customization, and implementation of distribution systems to meet diverse industrial and infrastructure needs.

7.1. OPPORTUNITIES IN INDIA'S POWER DISTRIBUTION AND CSS MARKET

Opportunities in the Power Distribution Panels & Compact Substations (CSS) Market
Valued at **INR 144.5 thousand crore in FY25** and projected to reach **INR 209.1 thousand crore by FY30F**, this market offers strong EPC potential in the **supply, installation, and integration of HT/LT panels and compact substations** across utilities, renewables, and industrial projects.

Growing **electrification, EV infrastructure, and smart city programs** are driving demand for **modular, automated distribution systems**, creating steady opportunities for EPC players in **turnkey execution and retrofitting** works.

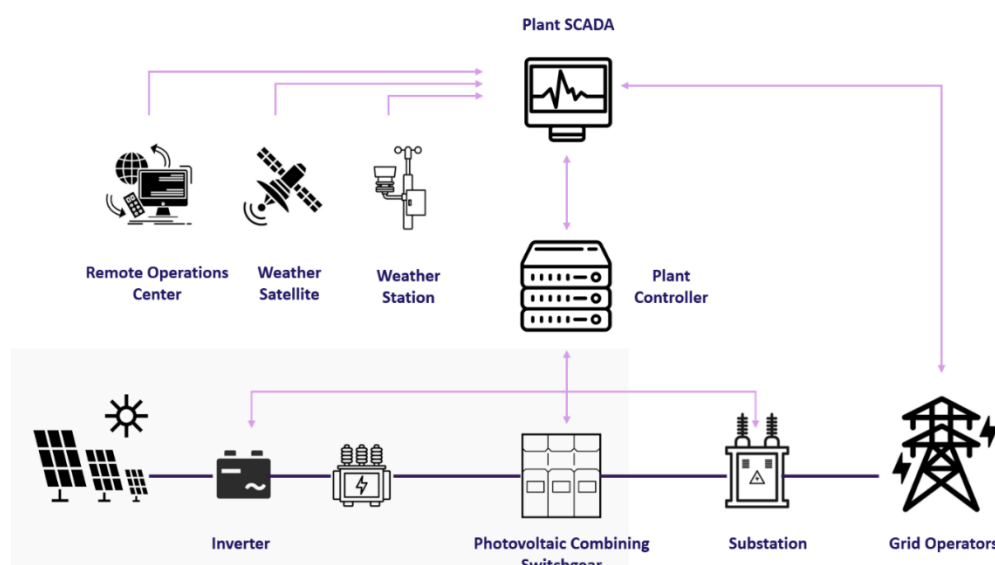
8. INDIA'S SCADA AND AUTOMATION SYSTEMS MARKET ANALYSIS

8.1. INDIA'S SCADA AND AUTOMATION SYSTEMS MARKET OVERVIEW

The **SCADA & Automation Systems Market** is defined as the domestic revenue generated from the design, supply, execution, and commissioning of SCADA and automation systems across electrical networks, water and wastewater infrastructure, power projects, and railway installations in India. The market is segmented by end-use industry (railways, water, power distribution), scope of work (design only, execution + commissioning), and system type.

Within India's automation landscape, **SCADA** is a specialized segment focused on **supervisory control, monitoring, and data acquisition** across distributed assets. It is majorly used in industries where **remote visibility** and **system reliability** are critical. **SCADA systems enable centralized control of field equipment, real-time data logging, fault detection, and performance optimization** making them essential for managing complex, large-scale infrastructure. This indicates that **SCADA remains a strategic enabler** for managing essential infrastructure.

Figure 8-1: SCADA Functioning in a Power Plant



Source: Ken Research Analysis

8.2. INDIA SCADA AND AUTOMATION SYSTEMS MARKET SIZE, FY20-FY30F

The Indian SCADA and Automation Systems market has demonstrated steady growth over the last five years, expanding from **INR 5.1 thousand crore in FY20** to **INR 7.7 thousand crore in FY25**, reflecting a **CAGR of 8.6%**. This growth momentum is expected to accelerate over the medium term, with the market projected to reach **INR 12.0 thousand crore by FY30F**, registering a **CAGR of 9.3% between FY25–FY30F**.

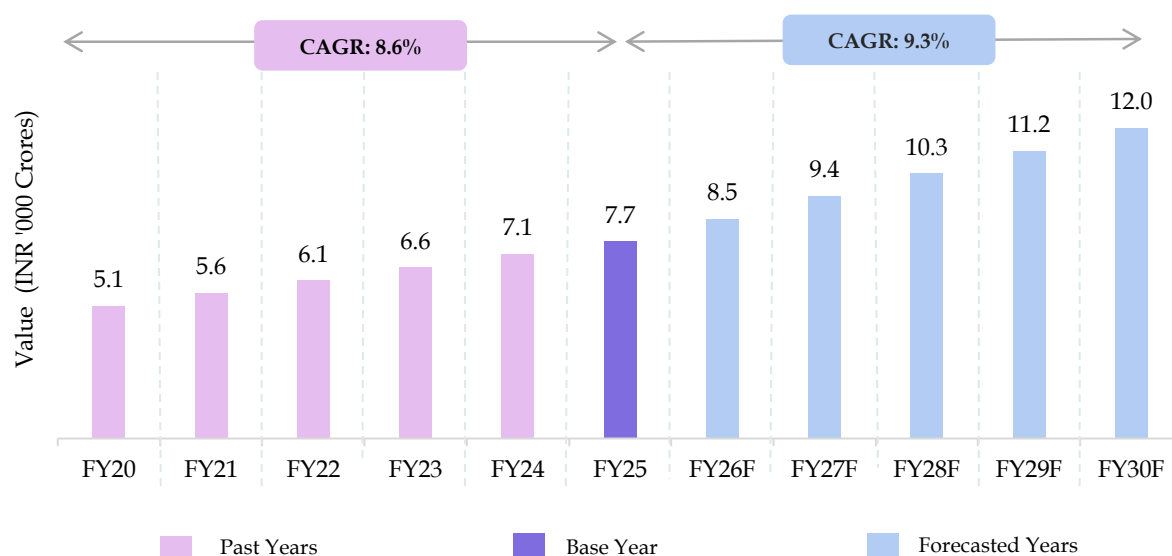
This upward trajectory is being driven by several structural factors:

- **Infrastructure modernization** – Increasing investments in railways, power T&D, water supply, and urban infrastructure are creating demand for advanced automation and monitoring systems.
- **Energy transition & renewables** – Expansion of renewable energy capacity and integration into the grid

requires sophisticated SCADA systems for real-time monitoring, balancing, and optimization.

- **Digital transformation & Industry 4.0 adoption** – Manufacturing, oil & gas, and process industries are scaling up automation to enhance productivity, reduce downtime, and improve safety.
- **Government-led initiatives** – Programs such as Digital India, Smart Cities Mission, and Make in India are fostering adoption of automation technologies across sectors.
- **O&M efficiency needs** – Utilities and industrial operators are increasingly relying on SCADA to lower operating costs and extend asset lifecycles.

Figure 8-2: India's SCADA and Automation Market Size in Terms of Value (in INR '000 Crore), FY20-FY30F



Source: Ken Research Analysis

Note: 1: FY represents Financial Year (April-March)

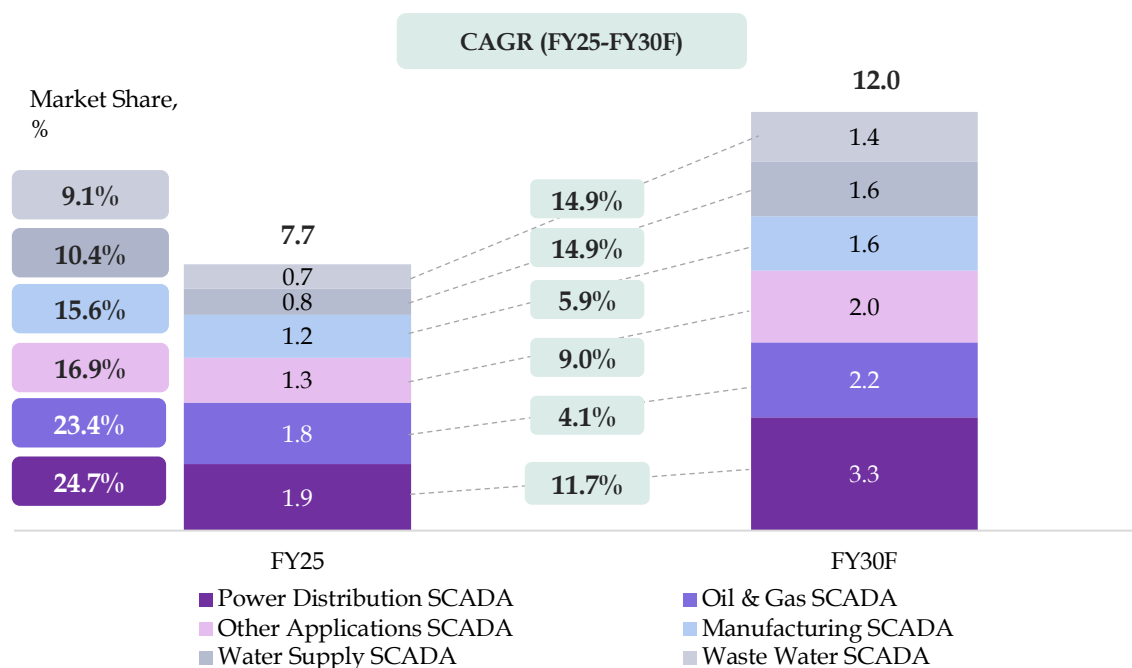
Note 2: F represents Forecasted figures

8.3. INDIA SCADA AND AUTOMATION SYSTEMS MARKET SEGMENTATION, FY25- FY30F

India's SCADA and Automation Systems Market Segmentation by Application, FY25-FY30F:

The Indian SCADA and Automation Systems market is projected to grow from **INR 7.7 thousand crore in FY25 to INR 12.0 thousand crore by FY30F**, at a CAGR of 9.3%. Growth is driven by modernization of utilities, integration of renewables, and rising demand for smart monitoring in water infrastructure.

- **Power Distribution SCADA** remains the largest application, supported by grid modernization, renewable integration, and distribution efficiency programs. By FY30F, it will consolidate leadership with strong double-digit growth.
- **Water Supply SCADA** is emerging as a high-growth segment, expanding rapidly with Jal Jeevan Mission rollouts, 24×7 supply initiatives, and the need for leakage detection and continuous monitoring.
- **Wastewater SCADA** is the fastest-growing segment, benefitting from large-scale deployment of municipal STPs, ETPs, and recycling plants under urban infrastructure programs.
- **Oil & Gas and Manufacturing SCADA**, while sizeable today, are expected to see comparatively slower growth due to tapering fresh CAPEX. These segments are increasingly sustained by O&M and digital upgrades rather than new installations.

Figure 8-3: India's SCADA and Automation Systems Market Segmentation (INR '000 crore) by Application, FY25 & FY30F

Source: Ken Research Analysis

Note: F represents Forecasted figures

Power remains the backbone of India's SCADA market, but water and wastewater applications are set to outpace traditional industries, reflecting the country's push towards sustainability and urban service reliability.

8.4. SCADA AND ITS INTEGRATION ACROSS PLATFORMS

Potential transformation of Traditional SCADA with IoT Integration

Integrating IoT devices with SCADA allows **real-time data acquisition** from sensors, meters, and field devices. It enables **remote monitoring, predictive maintenance, process automation, and security enhancements** by connecting edge devices directly to the control system.

Examples: **Siemens** uses IoT-enabled SCADA for smart grids and manufacturing automation.

Potential transformation of Traditional SCADA with GIS Integration

Combines spatial visualization from Geographic Information Systems with SCADA's real-time data to map substations, transformers, pipelines, and feeders.

Examples: **Reliance Energy** is utilizing SCADA with GIS integration to capture network changes on day-to day basis.

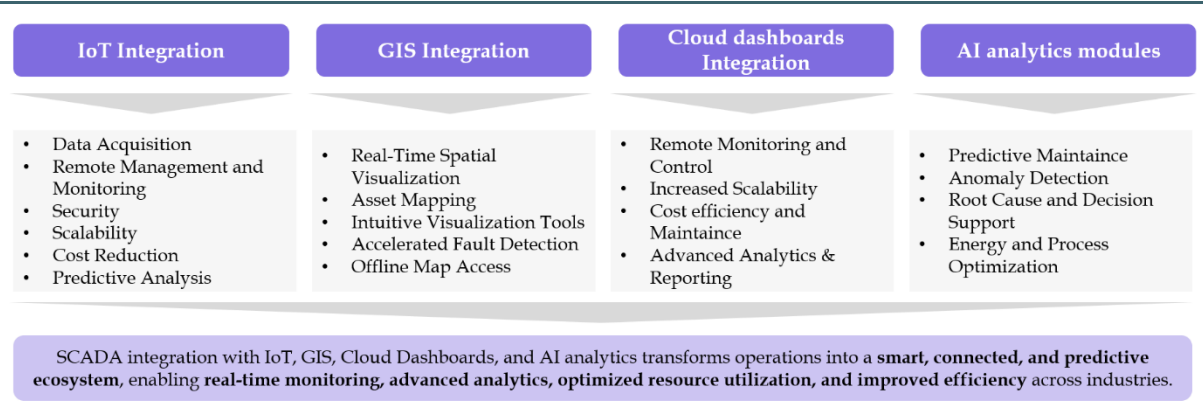
Integration of SCADA with Cloud Dashboards

Integrating SCADA with cloud dashboards enables remote monitoring and control through web-based interfaces. It supports scalability, advanced analytics, automated reporting, and reduced IT costs, making SCADA data accessible anywhere. Examples: **Bharat Light & Power** uses SCADA integrated with Google Cloud for remote monitoring and AI analytics of renewable energy operations, while Indian IT firms like **Wipro** and **Tata Consultancy Services** provide cloud-based SCADA integration solutions for industrial and utility sectors.

Industry transportation SCADA and AI Analytics

Embedding AI-powered analytics into SCADA enables predictive maintenance, anomaly detection, root cause analysis, and process optimization. AI models analyze historical and real-time SCADA data to provide decision support and performance optimization.

Figure 8-4: Benefits of Integration of SCADA across Platforms



Source: Copadata, Ken Research Analysis

8.5. OPPORTUNITIES IN INDIA’S SCADA AND AUTOMATION MARKET

Valued at **INR 7.7 thousand crore in FY25** and projected to reach **INR 12.0 thousand crore by FY30F**, the market offers growing opportunities for EPC players in **deployment and integration of SCADA systems** across power, water, and industrial assets.

With utilities adopting **IoT, cloud, and AI-based analytics**, EPCs with automation capabilities can capture value in **grid modernization, renewable integration, and smart water networks**, supported by **government-led digitalization programs** driving steady automation-linked EPC demand.

9. TRENDS AND DEVELOPMENTS

9.1. INDIA'S RAILWAY EPC MARKET

Increased Government Budget Pushing for Railway Network Expansion

The capital outlay supports critical areas such as increase in track length, S&T. In parallel, Ministry of Railways also plans to invest INR 16.7 lakh crore by 2031 toward freight corridors, electrification, high-speed rail lines, and the redevelopment of over 1,300 stations (Source: Economic Times). These efforts are being coordinated under the PM Gati Shakti National Master Plan, which has identified 434 railway projects across key logistics corridors (Source: Press Information Bureau Research Unit). This increases the EPC opportunities for players across the market. The planned investments and infrastructure expansion are driving demand for electrification to enable faster, energy-efficient, and sustainable rail operations across freight corridors, high-speed lines, and redeveloped stations.

A few instances can be seen through the targeted capital allocations for FY26 as outlined below:

Table 9-1: Union Budget key allocations within Ministry of Railways in INR crore, FY26

Purpose/Scheme	FY26 Fund Allocation in INR crore
New Railway Lines	32,235
Doubling and multi-tracking	32,000
Gauge Conversion	4,550
Broad-Gauge Electrification	6,500
Northeast Rail Connectivity Mission	10,440
High-Speed Rail (NHSRCL)	19,000
Dedicated Freight Corridors (<i>nearing completion</i>)	500

Source: Union Budget

Rising Focus on Specialized Projects increase demand for more contractual work

Indian Railways are shifting towards specialized projects such as tunnelling in difficult terrain, long-span bridges, high-speed corridors, and remote-region rail links. This transformation is fueled by modernization goals, urban transport demand, and requirement of strategic connectivity. The government has approved 12 major projects in the region, spanning 777 km with an investment of INR 69,342 crore. As of March 2025, 278 km has been commissioned, with INR 41,676 crore spent (Source: National Infrastructure Pipeline).

These projects require extensive support infrastructure, including transmission and distribution systems, overhead electrification (OHE), warehousing, and depots and more **creating strong growth opportunities for component manufacturers and infrastructure contractors**. Some of the key niche and specialized railways projects in India are:

Table 9-2: Key Niche and Specialized Railway EPC Projects in India

<u>High Speed Rail Corridors</u>	
Mumbai–Ahmedabad High Speed Rail	• 508 km bullet train corridor with 300 km+ viaducts and tunnels
Varanasi-Howrah High-Speed Rail Corridor	• 760 km route, expected to be completed by mid-2027
Delhi Amritsar High-Speed Rail Corridor	• 465 km route, at a speed of 350 km/hr
<u>Dedicated Freight Corridors</u>	
Eastern Dedicated Freight Corridor	• Route length of 1856 km consists of two distinct segments: • an electrified double-track segment of 1409 km between West Bengal & Uttar Pradesh • an electrified single-track segment of 447 km between state of Punjab, Haryana and Uttar Pradesh
Western Dedicated Freight Corridor	• 1,506 km, JNPT–Dadri; double-stack containerized freight
<u>Difficult Terrains / Strategic Projects</u>	

USBRL (Jammu-Baramulla Line)	• 272 km project, High-altitude tunnels & bridges, including world-high Chenab Bridge
Bairabi-Sairang (Mizoram)	• 51 km line with 48 tunnels, 55 major + 87 minor bridges
Rishikesh-Karnaprayag Rail Link	• 125 km Himalayan alignment with 16 tunnels

Source: Dedicated Freight Corridor Corporation of India Limited, Ministry of Railways

Increasing Private Participation through Schemes like Station Re-development is unlocking EPC Growth

Amrit Bharat Station Scheme (ABSS) is transforming over 1,300 stations with modern facilities, cultural integration, and urban connectivity improvements. While initially positioned as a PPP-driven initiative, limited private interest shifted the execution model towards EPC contracts, creating a pipeline of nearly INR 30,000 crore worth of opportunities for EPC players over the next two years (Source: ICRA). A select set of stations, including Rani Kamalapati, Pune, and Chennai Central, are being developed under the PPP model, but the bulk of the scheme relies on EPC, ensuring steady private participation through contracts while keeping investor risk minimal. **The infusion of private capital in redevelopment projects further expands opportunities for EPC players, enabling them to capture value in both execution and ancillary development.**

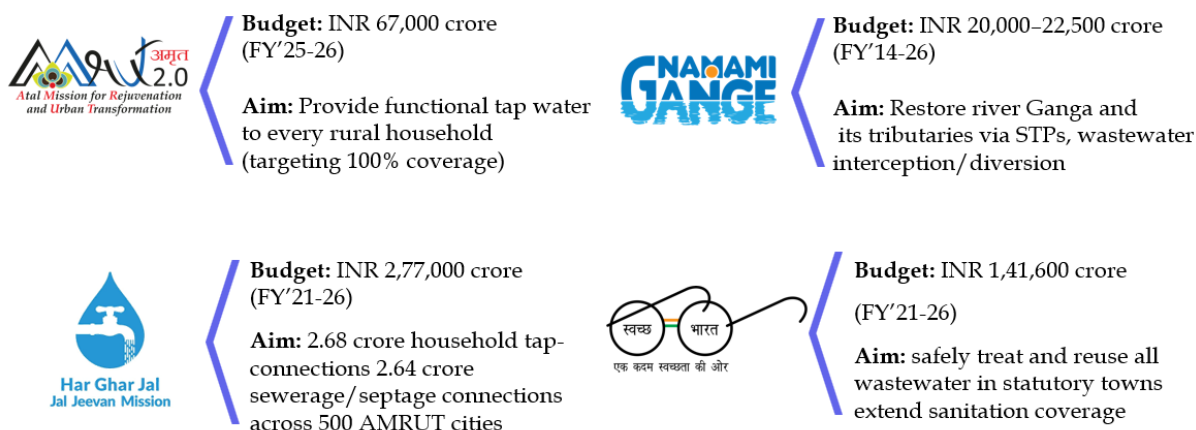
Urbanization and City Expansion Fueling Rail Infrastructure Demand

India's urban share is expected to rise from ~30% to 40–50% over the next 20–25 years, with 60+ million-plus cities contributing ~70% of GDP—a shift that concentrates travel demand and justifies sustained capex in metros, suburban corridors, and station redevelopment. As large cities densify and Tier-2/3 cities scale up, EPC needs expand across civil works, OHE, and S&T packages for multimodal hubs and decongestion projects.

9.2. INDIA'S DRINKING WATER AND WASTE WATER TREATMENT MARKET

Rapid Increase in Policy driven Investments is Driving Growth of Public Water Infrastructure

Policy-driven investments are accelerating India's **water and wastewater treatment market**, boosting demand for **private players**. Flagship programs like **Jal Jeevan Mission, AMRUT 2.0, and Namami Gange** are driving large-scale projects in **pipelined water supply, STPs, river rejuvenation, and reuse**. Urban initiatives such as **Delhi's Yamuna cleanup, Bengaluru's INR 3,700 crore lake revival, and Vizag's INR 1,200 crore desalination plant** reflect rising investment. New **CPCB norms and reuse policies** are pushing adoption of advanced technologies including **tertiary treatment, decentralized STPs, desalination, and smart water systems**. **PPP models, green bonds, and multilateral funding** are also expanding private sector participation, driving **sustained growth in capital expenditure** across the sector.

Figure 9-1: Key Government Policies impacting Drinking Water and Waste Water Treatment Market

Source: Ministry of Housing & Urban Affairs

Government Policies combat Water Stress require appropriate infrastructure

Globally, water stress varies widely, with many regions in South America, Central Africa, and Northern Europe experiencing low to moderate levels of pressure on freshwater resources. Developed economies such as the United States, Canada, and much of Western Europe generally fall in the medium to low stress category, with only a few localised areas facing high scarcity. By contrast, **India falls into the “extremely high” stress bracket, with more than 80% of available renewable water resources already being withdrawn annually placing it among the most water-stressed nations worldwide and highlighting a challenge far more acute than that faced by most developed and other major developing countries.**

(Note: Low (<10%), Low-medium (10–20%), Medium-high (20–40%), High (40–80%), Extremely high (>80%).)

India faces acute urban and industrial water stress, with only 28% of urban wastewater currently treated, raising serious concerns over scarcity and pollution (Source: Centre for Science and Environment). To address this, the

Government of India has launched major initiatives backed by **significant investments** to expand **drinking water access**, build **sewage treatment plants**, and promote **wastewater recycling**.

These efforts are creating strong demand for **Engineering, Procurement & Construction (EPC)** players to design and execute large-scale infrastructure such as **treatment plants, recycling systems, smart water infrastructure, and pipeline networks**. As a result, **capital expenditure in water infrastructure is rising**, positioning EPC companies at the core of **India's push for sustainable water management**.

Emergence of Advanced Treatment Technologies drive EPC demand

Adoption of advanced treatment technologies membrane bioreactors (MBRs), advanced oxidation processes (AOPs), electrochemical systems and AI/IoT-enabled digital twins is accelerating as government and industrial actors upgrade facilities to meet tighter reuse and discharge standards. Further, this is accelerating demand across the **Drinking Water & Wastewater Treatment** market for **advanced treatment assets and its EPC**.

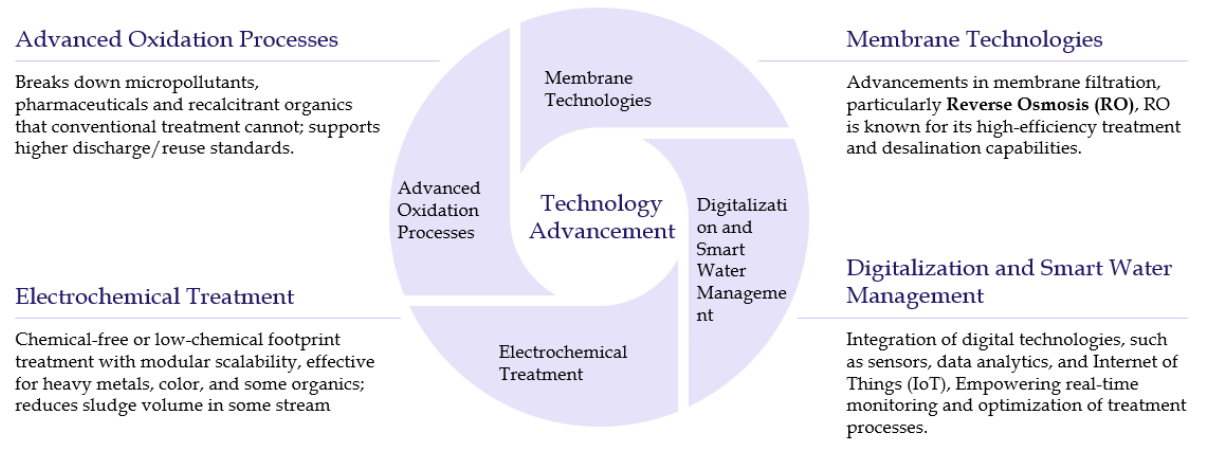
Stringent Environmental & Public Health Regulations

There is a persistent gap between sewage generated and treated across urban centers, driving regulatory urgency for new and upgraded STPs and ETPs. In response, state agencies and ULBs are accelerating project execution under AMRUT 2.0, which mandates SCADA-based monitoring and advanced automation in approved sewerage works. Simultaneously, updated OCEMS norms for real-time effluent tracking are pushing industries toward technology-led compliance—broadening EPC

demand across civil, electro-mechanical, and automation packages. (Source: Central Pollution Control Board)

Aging Infrastructure & Need for Upgrades
Legacy networks and under-capacity plants are being overhauled through mission funding: under AMRUT 2.0, approved projects already propose 4,814 MLD of new STP capacity, alongside city-level reforms targeting NRW <20% – both of which require pipeline rehabilitation, DMA creation, pressure management, energy-efficient pumping, and SCADA integration.

Figure 9-2: Major Technological Advancements in Drinking Water & Wastewater Treatment



Source: Ministry of Housing & Urban Affairs

9.1. INDIA TRANSMISSION AND DISTRIBUTION MARKET

Rising Power Demand & Industrial Load Growth Driving Network Expansion

India’s electricity demand continues to surge, with peak demand touching **250 GW in FY2024–25** and expected to reach nearly **270 GW by FY2026**, driven by industrial

expansion, data centers, and urbanization (Source: Ministry of Power). This load growth is prompting utilities to expand **400 kV and 765 kV networks**, strengthen substations, and develop new interconnection corridors. States such as Karnataka and Maharashtra are upgrading to higher-voltage systems to support large industrial parks and technology clusters. Public utilities like **POWERGRID** and private developers such as **Adani Energy Solutions** are scaling up capital expenditure to meet this demand, creating a sustained pipeline for transmission EPC firms (Source: Adani Energy Solutions).

Technology Shift and Reconductoring Enabling Faster Capacity Addition

Right-of-way constraints and urban congestion are leading utilities to adopt **High-Temperature Low-Sag (HTLS) reconductoring** and compact tower technologies to boost corridor capacity without building new lines (Source: CEA). Reconductoring can effectively double line capacity within shorter project cycles, making it an efficient solution for dense urban and industrial zones. Indian manufacturers like **APAR Industries** and **Sterlite Power** are executing major reconductoring and uprating projects, offering faster commissioning and minimal downtime. This shift favors EPC firms equipped for **live-line upgrades, bay expansions, and compact tower deployment**, adding a specialized niche within India's transmission modernization program (Source: IEEMA).

Electric Mobility Expansion Accelerating Urban Distribution Reinforcement

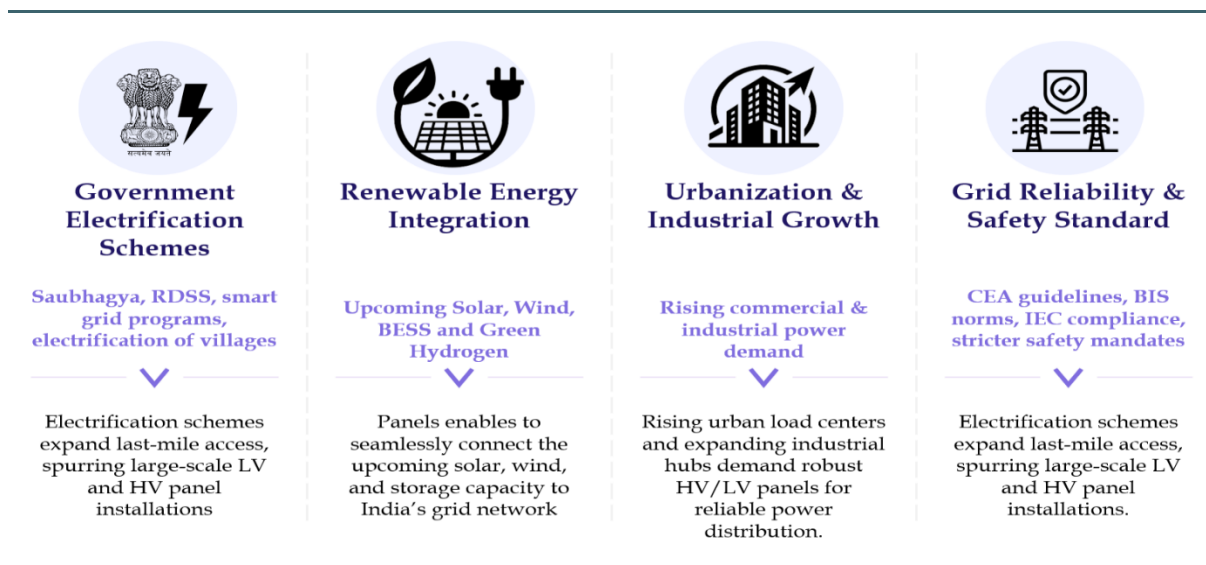
India's transition to e-mobility is adding new layers of demand to urban power networks. Over **2 million electric vehicles** were sold in FY2024–25, while the number of public charging stations approached **30,000** nationwide

(Source: NITI Aayog). The rollout of the **PM E-DRIVE** program, providing dedicated funding for public charging infrastructure, is accelerating the creation of EV corridors across major cities and highways. DISCOMs such as **DHBVN and BESCOM** are strengthening **11-33 kV feeders**, upgrading distribution transformers, and adopting **smart load management** systems to manage evening charging peaks. This transition is generating recurring EPC opportunities in **urban feeder reinforcement, automation, and charging substation deployment** (Source: Ministry of Power).

9.2. INDIA'S POWER DISTRIBUTION PANEL AND COMPACT SUBSTATION MARKET

Expanding Access and Grid Upgradation are Driving Demand for HV and LV Power Distribution Panels

As shown in the figure below, a combination of government schemes, renewable integration, urbanization, automation, and stricter reliability norms is reshaping India's power sector. Together, these forces are accelerating the need for HV and LV distribution panels, which are vital for managing new connections, enhancing grid stability, and enabling modernized power systems.

Figure 9-3: Factors Driving Demand for Power Distribution Panels in India

Source: Ken Research Analysis

Note: RDSS: Revamped Distribution Sector Scheme

Industrial Expansion Accelerates Demand for HV & LV Distribution Panels

India's industrial footprint is rapidly expanding, creating significant downstream demand for power distribution infrastructure.

- In FY24 alone, the country added **6,727 new factories**, the highest in 13 years (Source: Moneycontrol).
- In parallel, the **National Industrial Corridor Development Programme** is advancing **12 new smart industrial city nodes** to support large-scale manufacturing growth (Source: IBEF, Invest India).

Together, these trends are translating into higher electricity requirements within industrial hubs, reinforcing the need for HV and LV distribution panels to ensure reliable, efficient, and scalable power supply.

Technological Advancement

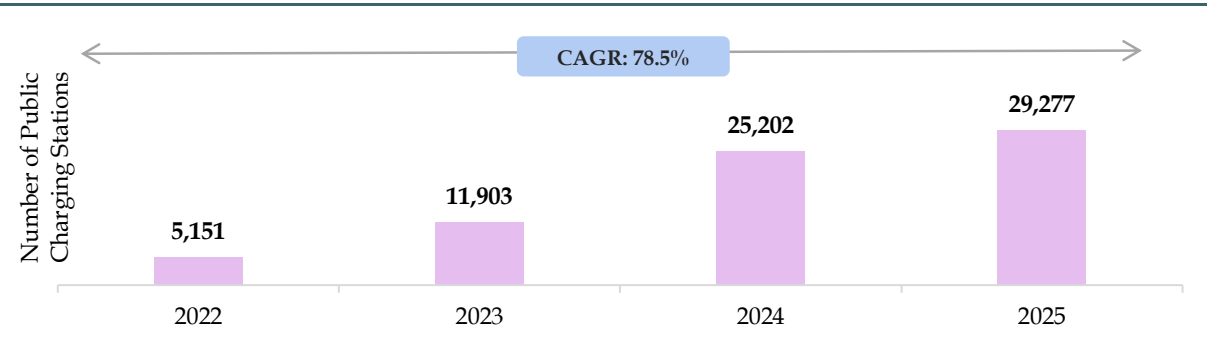
With 20.33 crore smart meters sanctioned and 2.41 crore installed as of July 15, 2025, DISCOMs must upgrade

downstream panels, RMUs, protection gear, and compact substations to support AMI/SCADA integration. RDSS funding explicitly supports SCADA, feeder segregation, and distribution infrastructure, heightening EPC demand for modern panels and switchgear. Compact Secondary Substations (CSS) factory-assembled, type-tested units combining MV and LV components are gaining adoption for dense urban and renewable-heavy zones.

EV Adoption and Charging Infrastructure Driving Panel Demand

India's electric vehicle market is accelerating, with **electric four-wheelers registering a 15.68% year-on-year growth in FY 2024–25**, reaching 115,800 units (Source: Vahan Dashboard). Simultaneously, public EV charging infrastructure **has surged 5X from 2022 to early 2025**, growing from **5,151 stations to 29,277**, at a **~78.5% CAGR** over three years (Source: Ministry of Heavy Industries). The combined rise of EV adoption and charging infrastructure is increasing load demand in urban centers, transport depots, and commercial facilities, **fueling the need for robust HV and LV distribution panels to ensure safe and reliable power supply**.

Figure 9-4: Public EV Charging Stations in India, as on January 2026



Source: Ministry of Heavy Industries, Ken Research Analysis

Distribution Reforms and Substation Capacity Enhancement

The **Revamped Distribution Sector Scheme (RDSS)**, launched by the Government of India with a total outlay of **INR 3.03 lakh crore till FY26**, is driving significant investment in distribution infrastructure. A key focus of the scheme is **strengthening and augmenting substation capacity**, alongside modernization of distribution networks to reduce AT&C losses and improve reliability (Source: Ministry of Power). Within this push, technologies such as **Compact Substations (CSS)** are gaining traction as utilities look for space-efficient and modular solutions that can be deployed in dense urban areas and rapidly growing industrial zones. The emphasis on capacity enhancement under RDSS is expected to directly spur demand for HV and LV panels, as well as allied equipment such as CSS.

9.3. INDIA SCADA AND AUTOMATION SYSTEMS MARKET

Policies are Driving SCADA Integration in India's Utility and Industrial Sectors

India's push for SCADA integration is being led by key policies like the **Smart Cities Mission**, **AMRUT 2.0**, and the **Revamped Distribution Sector Scheme (RDSS)**. These initiatives are embedding SCADA into urban infrastructure to improve reliability, efficiency, and real-time control.

In **Smart Cities**, SCADA powers **Integrated Command and Control Centers (ICCCs)** for monitoring utilities like water, power, and traffic. **AMRUT 2.0** mandates SCADA for water network management and leak detection. Further, **RDSS** projects like in **Noida** have earmarked **INR 396 crore** for SCADA-backed grid automation.

As utilities expand, SCADA systems are becoming vital for deployment in dense urban areas. On the **industrial front**,

sectors like **steel**, **cement**, and **chemicals** are adopting SCADA for automation, energy efficiency, and predictive maintenance aligning with India's digitalization goals. Together, these efforts are making SCADA central to managing urban utilities and optimizing industrial operations.

Table 9-3: Schemes Driving need of SCADA Systems in Utilities

Scheme Name	SCADA Requirement Description
National Smart Grid Mission (NSGM)	SCADA is essential for grid automation, real-time load dispatch, fault detection, and renewable integration.
Revamped Distribution Sector Scheme (RDSS) – Noida	SCADA-backed automation is a core component; INR 396 crore allocated specifically for SCADA systems.
SECI – Wind, Solar, Hybrid Projects	SCADA is required for plant performance monitoring, wind forecasting, and grid synchronization.
Urban Water Distribution – AMRUT Mission	SCADA is mandated for monitoring flow, pressure, reservoir levels, and leak detection in water supply systems.

Source: Union Budget

Remote / Unmanned Operations Drives Demand

India's utilities are shifting to remote-controlled and unmanned substations to cut downtime and manpower costs. CEA's technical standards and the RDSS scheme promote centralized SCADA/DMS control and automation of field assets, with grant support up to 90% for such projects. State utilities like Andhra Pradesh and Gujarat have begun converting substations to fully remote operation, creating sustained EPC demand for control systems, sensors, and secure communication networks. (Source: Ministry of Power)

IoT and AI as Enablers of SCADA Growth in India

The use of **IoT** and **AI** is gradually expanding in SCADA and automation systems across India. These technologies are being applied to improve monitoring, control, and maintenance in both urban infrastructure and industrial operations.

Companies such as **Britannia Industries** and **Itanta** have implemented IoT-based SCADA across production lines. These systems enable real-time tracking of production metrics and automate functionalities such as receiving of fault alerts, changing production speeds etc.

While IoT adoption is already widespread, **AI integration is still emerging**. Companies such as **Tata Steel** and **Bharat Forge** have started using AI for tasks like predictive maintenance and process optimization. As more industries explore these tools, AI is expected to become a standard part of SCADA systems for improving operational efficiency and reliability.

10. THREATS AND CHALLENGES

10.1. CHALLENGES IN THE MARKET

The challenges that EPC players face across the 4 markets, have been highlighted down below:

Table 10-1: Challenges faced by EPC players across the 5 target markets

India's Railway EPC Market		
Challenges	Impact	Description
Payment Delays Straining Working Capital and Slowing Project Execution		Financial challenges stem from payment delays and high bank guarantees that lock up working capital. Indian Railways contracts allow monthly billing based on verified work, but no advance payments and multi-level approvals often cause significant payment lags. This strains smaller players in the market.
Design and Engineering Risks		Frequent design revisions, incomplete surveys, and delayed approvals from rail authorities lead to rework, idle resources, and extended timelines.
Price Volatility of Steel, Cement, and Copper Inputs		Sharp fluctuations in key input prices erode margins under fixed-price contracts despite escalation clauses, as costs of steel, copper, cement, and aggregates often surge unpredictably, straining cash flows and project profitability.

Restricted Site Access on Live Tracks Complicating Execution and Civil Works in Dense Corridors		Execution on live tracks is constrained by short block windows (2–3 hours), requiring rapid work under tight safety protocols. In dense corridors, limited space, underground utilities, and encroachments further hinder civil works and resource coordination. This ends up delaying projects and results in extended timelines.
Elongated Supply chain		Multiple subcontractors and long lead times for fabricated and electrical components disrupt project schedules and increase logistics complexity, as coordination across suppliers, testing agencies, and transport networks often leads to delays, rework, and cost overruns in large, multi-location railway EPC projects.

India’s Drinking Water and Waste Water Market

Challenge	Impact	Description
Operational Inefficiencies		Skills shortages, outdated assets, and inefficient processes reduce uptime and compliance, increasing project risks and execution costs for EPC players.
High Capex and Maintenance Costs		Building treatment plants involves heavy upfront investment in pumps, filtration systems, pipelines, and compliance infrastructure. Ongoing O&M expenses such as energy-intensive operations, skilled staffing, and equipment upgrades strain financial viability.
Residuals Handling & Disposal		Sludge, screenings, and RO concentrate need compliant dewatering, transport, and disposal; limited landfill/incineration options and permits create bottlenecks and shutdown risk.

India’s Power Distribution Panels and Compact Substation Market

Challenge	Impact	Description
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Fragmented and Unorganized Market		Over 70% of players are small-scale and unorganized, this causes pricing pressure, irrespective of the fact of low quality by smaller players and what supply chain inefficiencies they may have.
Dependence on Tender-Based Procurement		A large share of demand comes from government and utility projects, making the market heavily tender-driven. This results in long approval cycles, high competition, and margin pressures, impacting revenue predictability and timelines.
Excessive Expenditure on Heat Management		High temperatures and compact enclosures cause overheating and equipment failure; requires proper ventilation and derating. This causes inefficiencies and reduces profitability.
Cost Pressures		Escalating prices of copper, steel, and other metals, combined with additional expenses for product certification, safety testing, and site-specific customization, significantly squeeze margins under competitive bidding environments.
Safety and Reliability Risks		Arc-flash incidents, poor earthing practices, and water ingress remain persistent risks that can lead to major equipment failures, power outages, and worker injuries. Ensuring robust design, proper installation, and strict adherence to IEC and IS standards become expensive.

India's Transmission and Distribution Market

Challenge	Impact	Description
Dependence on Tender-Based Procurement		Most T&D projects are utility-led, involving lengthy bid cycles, intense price competition, and thin margins. Delayed awards and strict qualification norms further strain cash flows and limit flexibility for EPC players.

Weak Financial Health of DISCOMs		Chronic payment delays, mounting subsidy backlogs, and unresolved tariff gaps strain contractor cash flows and escalate financing costs. In several cases, delayed clearances and disputes over project billing or variation orders have led contractors to pursue prolonged legal or arbitration proceedings, tying up working capital and disrupting execution timelines.
Right-of-Way (RoW) & Clearance Delays		Land disputes and slow environmental approvals frequently stall transmission projects and extend timelines.

India’s SCADA and Automation Systems Market

Challenge	Impact	Description
High Capital Cost and Integration Complexity		It requires upfront investment in hardware, software licenses, cybersecurity frameworks, and networking infrastructure. Integration with legacy systems adds technical challenges and cost burdens, slowing adoption.
Shortage of Skilled Technicians		SCADA deployment and maintenance demand specialized expertise in both operational technology (OT) and information technology (IT). A shortage of trained personnel delays implementation, increases dependency on third parties

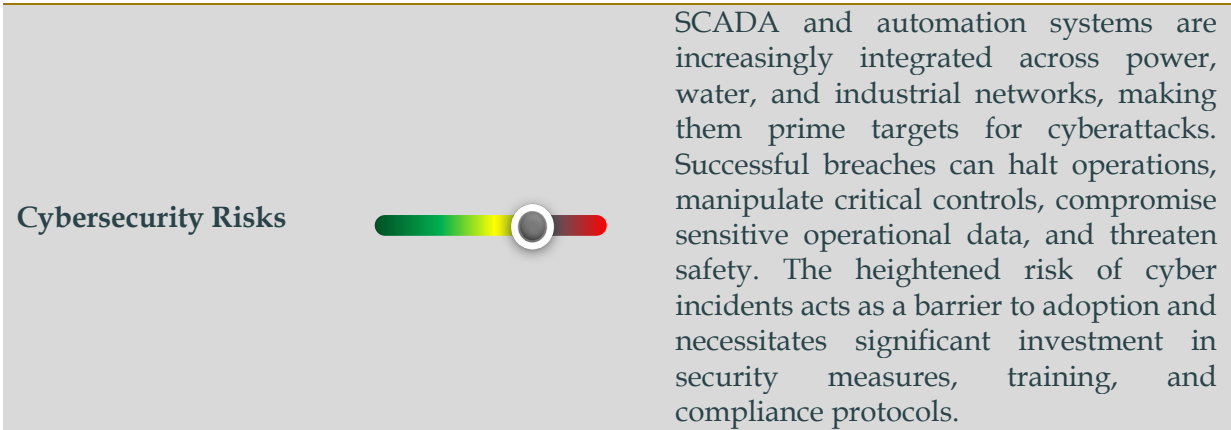
Source: Ken Research Analysis

10.2. THREATS IN THE MARKET

The threats that EPC players face across the 4 markets, have been highlighted down below:

Table 10-2: Threats impacting EPC players across the target markets

India's Power Distribution Panels and Compact Substation Market		
Threats	Impact	Description
Price Volatility of Key Raw Materials		Key raw materials like copper, aluminum, and steel are highly price-sensitive and subject to global market fluctuations (e.g., copper showing ~22% YoY volatility). Such unpredictability complicates cost forecasting for EPC contractors and panel manufacturers. Projects initiated under one cost assumption may quickly become unprofitable if raw material prices spike, discouraging investment and affecting competitive bidding.
India's Drinking Water and Waste Water Market		
Threats	Impact	Description
Delays in Land Acquisition & Clearances		Development of water and wastewater treatment plants is highly dependent on timely land allocation and environmental approvals. Local disputes, lack of clarity in regulatory processes, or extended permit reviews can stall construction, increase financing costs, and disrupt project timelines. Such delays also limit the ability to meet growing urban water demand and impact operational scaling.
India's SCADA and Automation Market		
Threats	Impact	Description



Source: Ken Research Analysis

11. REGULATORY LANDSCAPE

11.1. REGULATORY LANDSCAPE OF INDIA'S RAILWAY EPC MARKET

The landscape for Railway EPC is governed by key listed regulations: -

The Railways Act, 1989

Enacted in 1989 and amended multiple times, this Act provides the legal foundation for railway development, safety, and land use. It empowers authorities to acquire land, regulate construction, and oversee operations. Compliance requires executing works strictly as per notified procedures for acquisition, design, and safety, ensuring no deviation from statutory railway standards.

Railway Board Circulars and Manuals

Issued periodically and updated over time, these circulars and manuals define technical standards, engineering practices, and SOPs for civil, electrical, and mechanical works. Compliance entails incorporating the latest specifications into planning, design, and execution to meet operational and safety benchmarks.

Indian Railways General Conditions of Contract (GCC)

Launched as a standard contracting framework and updated regularly, the GCC prescribes obligations, timelines, penalties, and quality norms for executing works. Alignment involves adhering to contract timelines, quality standards, and documentation requirements to avoid disputes and penalties.

Indian Railways Works Manual / Unified SSR

Introduced to standardize estimation and execution of works, these documents lay down cost norms and procedural workflows. Compliance includes using prescribed rates, preparing accurate estimates, and following established procedures for budgeting, approvals, and execution.

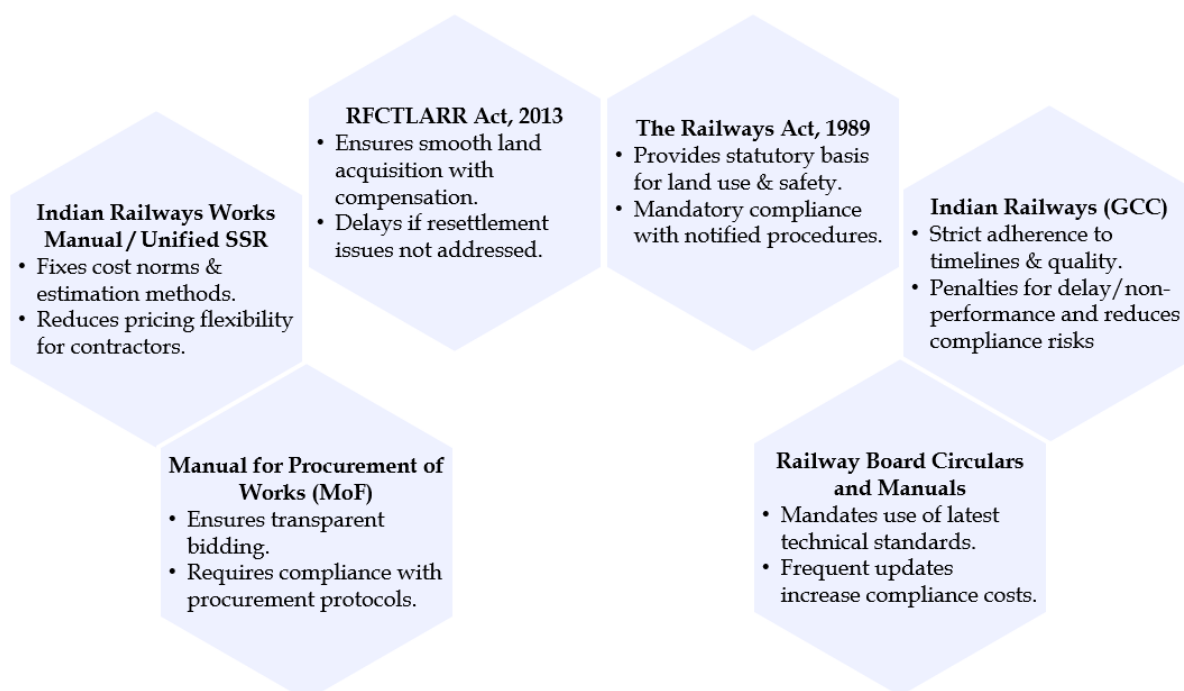
RFCTLARR (Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement) Act, 2013

It ensures fair compensation, transparency, and rehabilitation for landowners affected by public or private projects. It mandates social impact assessments, consent from landowners, and compensation, replacing the 1894 Land Acquisition Act.

Manual for Procurement of Works (MoF)

First issued by the Ministry of Finance in 2019 and updated in 2022, this manual sets rules for tendering, contracting, and execution of public works. Compliance requires transparent bidding, adherence to timelines, and following prescribed procurement protocols during all project stages.

These regulations ensure that railway EPC projects follow strict legal, technical, and procedural standards. They make compliance with land acquisition, safety, contracting, cost norms, and procurement rules an essential part of project execution.

Figure 11-1: Impact of Major Regulatory on Railway EPC Players

Source: Ken Research Analysis

11.2. REGULATORY LANDSCAPE OF INDIA'S DRINKING WATER AND WASTE WATER TREATMENT MARKET

The regulatory landscape for Drinking Water and Waste Water Treatment is governed by the following guidelines and acts:-

Water (Prevention and Control of Pollution) Act, 1974

This act established the legal framework for controlling water pollution through the creation of CPCB and SPCBs, requiring prior consent for any discharge into water bodies.

Environment (Protection) Act, 1986

A comprehensive law empowering the central government to set environmental standards, including for wastewater discharge, and enforce them nationwide.

CPCB Guidelines & MOEF&CC Notification (2017)

CPCB proposed STP standards in 2015, which became legally binding after MOEF&CC's 2017 notification, setting discharge norms for BOD and TSS based on city classification.

Central Ground Water Authority (CGWA) Guidelines

These guidelines regulate groundwater extraction, requiring industries and municipalities to obtain NOCs and follow sustainable usage practices.

National Water Policy, 2012

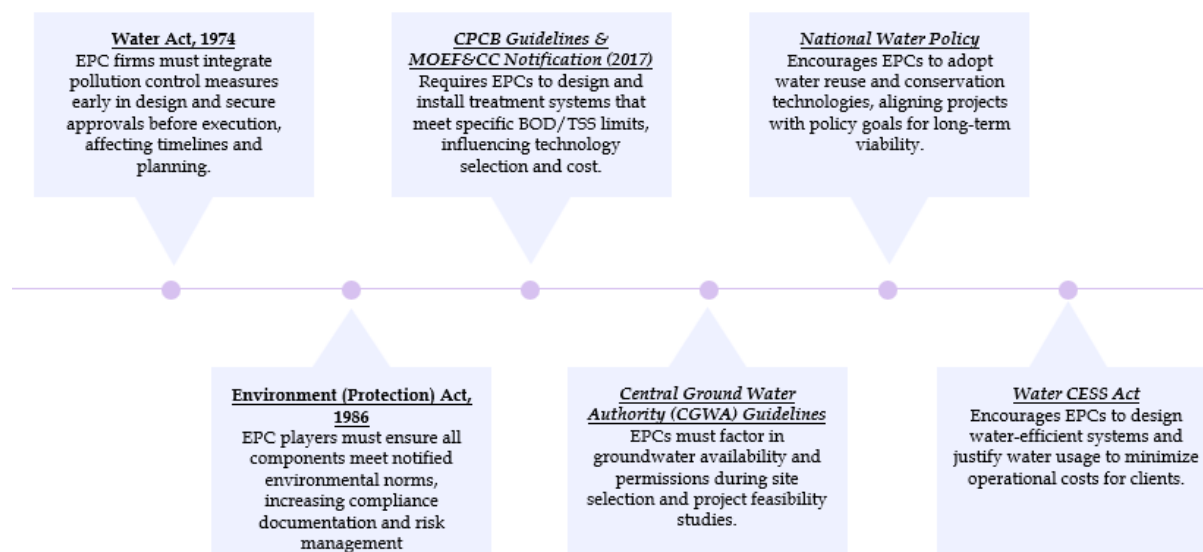
A strategic policy promoting integrated water resource management, prioritizing drinking water needs, and encouraging reuse of treated wastewater.

Water CESS Act (1977, amended 2003)

Allows authorities to levy charges on water consumption, especially by industries, to fund pollution control and water conservation efforts.

These regulations mandate effluent treatment, set discharge standards, and restrict groundwater use. The Water and Environment Acts, CPCB norms, and CGWA guidelines enforce compliance, while the Water CESS Act adds economic accountability. The National Water Policy further emphasizes reuse, making treatment and recycling essential parts of water management.

Figure 11-2: Impact of major regulations on Drinking Water and Wastewater EPC players



Source: Ken Research Analysis

11.1. REGULATORY LANDSCAPE OF INDIA'S TRANSMISSION AND DISTRIBUTION MARKET

The regulatory landscape for Transmission and Distribution infrastructure is governed by the following guidelines and acts:-

Transmission System Planning & Development Regulations, 2021

Establishes coordinated national and state-level grid planning linked to 500+ GW renewable targets. Ensures transparent tariff recovery and promotes competitive bidding (TBCB), boosting private EPC participation.

Rights of Way (RoW) Rules, 2016 (Amended 2023)

Standardizes compensation and clearance processes for overhead and underground lines. Reduces project delays and eases execution for transmission contractors.

Central Transmission Utility (CTU) & State Transmission Utility (STU) Regulations, 2020

Define responsibilities for grid planning, approvals, and inter-state coordination, improving efficiency in project rollout.

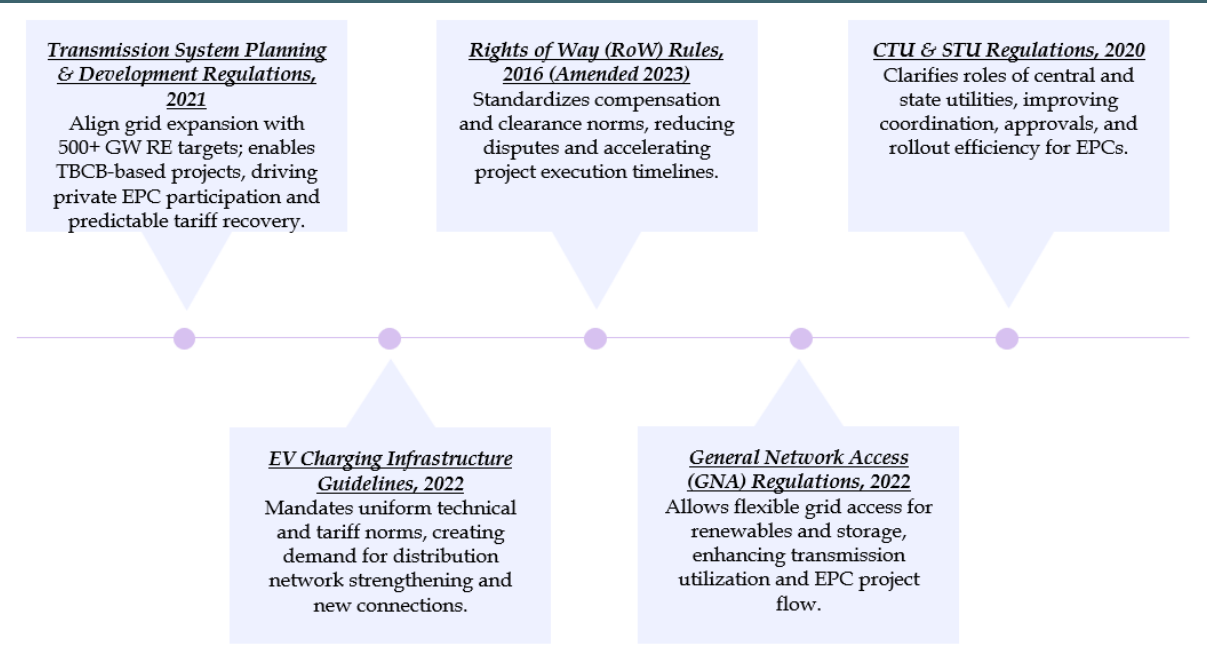
Electric Vehicle Charging Infrastructure Guidelines, 2022

Set technical and tariff standards for EV charging networks. Trigger distribution-level reinforcement and new connection demand.

General Network Access (GNA) Regulations, 2022

Redefines access to the national grid, allowing flexible and efficient capacity use for renewable and storage projects. Enhances transmission utilization and expands EPC opportunities in inter- and intra-state corridors.

Figure 11-3: Impact of major regulations on Transmission and Distribution sector players



Source: Ken Research Analysis

11.2. REGULATORY LANDSCAPE OF INDIA'S HT / LT DISTRIBUTION PANELS & COMPACT SUB-STATIONS MARKET

The landscape for HT/LT distribution panels and compact substations is anchored by:

Electricity Act, 2003

Parent law governing generation, transmission, distribution, and electrical safety across India. Provides statutory backing for compliance and empowers regulatory authorities.

CEA (Measures Relating to Safety and Electric Supply) Regulations, 2023

Prescribes mandatory safety norms for design, installation, operation, and maintenance of electrical systems to prevent hazards and ensure reliable supply.

CEA Technical Standards for Construction of Electrical Plants & Lines, 2022

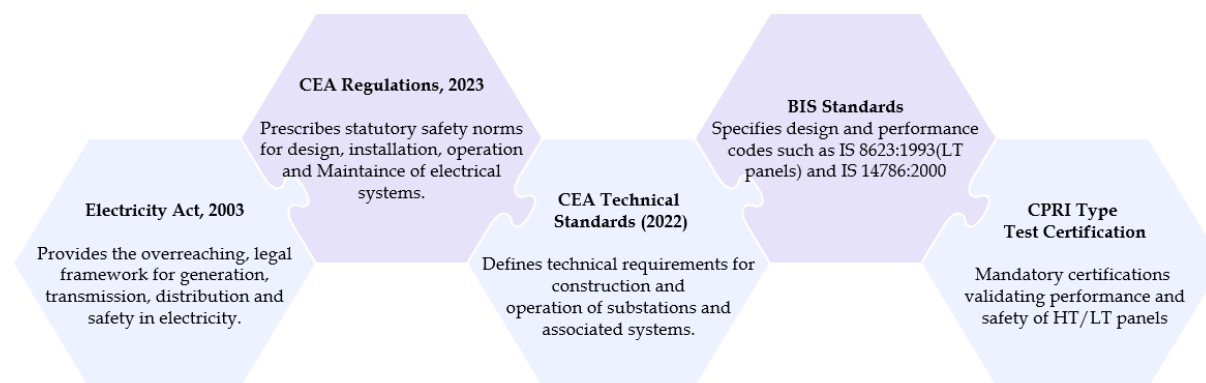
Defines technical requirements for construction and integration of substations, plants, and lines, ensuring uniformity and safe grid connectivity.

BIS Standards (IS 8623:1993, IS 14786:2000)

Indian Standards that set design and performance requirements for LT panels (IS 8623) and prefabricated substations (IS 14786).

CPRI Type Test Certification

Independent testing of panels and substations by CPRI/NABL labs to validate busbar ratings, thermal withstand, and safety performance.

Figure 11-4: Impact of Major Regulations on HT / LT Distribution Panels & Compact Sub-Stations EPC Players

Source: Ken Research Analysis

11.3. REGULATORY LANDSCAPE OF INDIA'S SCADA AND AUTOMATION SYSTEMS MARKET

The regulatory landscape for SCADA and Automation Systems is governed by the following guidelines and acts:-

Electricity Act, 2003

It is the legal backbone for power sector operations, grid standards, and system reliability. It mandates monitoring, scheduling, and dispatch systems – SCADA/DMS becomes a tool to comply.

CEA Safety & Grid Standards Regulations (2010, 2022)

Ensure reliable grid operations, data availability, and stability. It demands for real-time monitoring and data acquisition to maintain grid discipline.

CERC Communication System Regulations (2017)

Mandates communication infrastructure for real-time monitoring. SCADA systems must seamlessly integrate with Wide Area Measurement Systems (WAMS) and PMUs.

National Smart Grid Mission (NSGM), 2015

Modernize the grid through smart automation, demand management, and advanced metering and SCADA is at the core of smart grid rollouts.

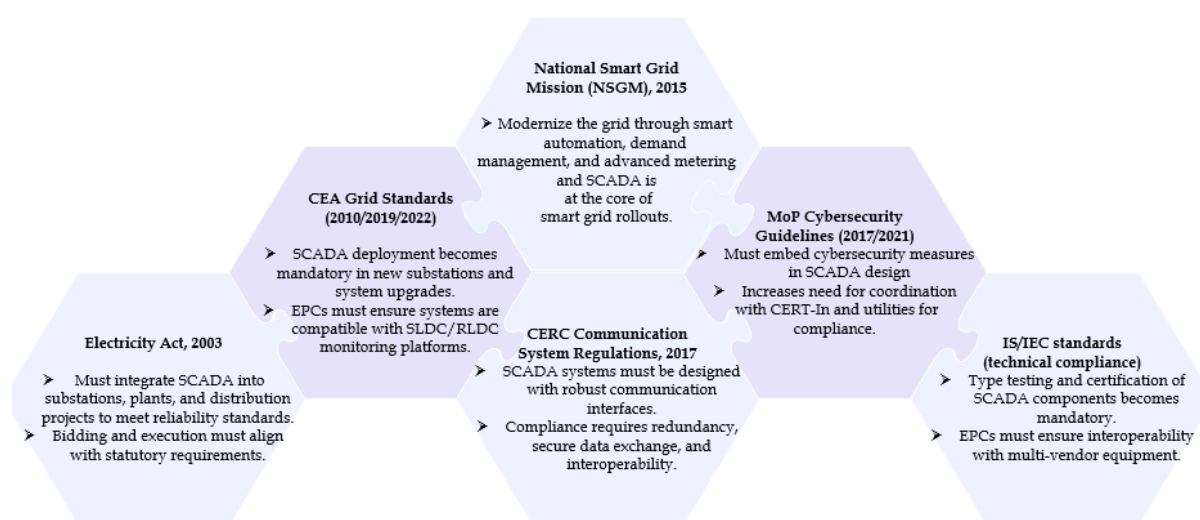
MoP Cybersecurity Guidelines (2017/2021)

It establish enforceable baseline rules for all power-sector entities (utilities, vendors, OEMs) to systematize cyber governance, IT/OT segregation, logging and monitoring, incident response, audits, and secure procurement.

IS/IEC standards (15953, 61850, 62351, etc.)

It comprises of communication protocols for substation automation, cybersecurity for power system control and SCADA. These regulations are essential for ensuring grid reliability, real-time monitoring, secure communication, and compliance with evolving smart grid standards in India.

Figure 11-5: Key impact of these regulations on SCADA and Automation systems EPC players



Source: Ken Research Analysis

12. GOVERNMENT INITIATIVES

Government-led infrastructure programs are reshaping India's core sectors, creating sustained opportunities for EPC players. In particular, flagship initiatives in railways and drinking water & wastewater management are driving large-scale demand for engineering, construction, and systems integration.

12.1. INITIATIVES DRIVING DEMAND FOR RAILWAY EPC PLAYERS

Major Government Initiatives Creating Demand for EPC Player are:-

- **Vande Bharat Scheme**
Flagship semi-high-speed train program targeting pan-India deployment of new-generation trainsets. It aims Manufacture and operates 200 Vande Bharat trains within three years, supported by dedicated maintenance depots, upgraded stations, and modern signaling. This drives large civil, mechanical, and electrical EPC contracts ~INR 30,000–40,000 crore for depots, electrification, and signaling system upgrades.
- **Rail Land Development Authority (RLDA) Projects**
RLDA oversees commercial redevelopment of railway land, stations, and colonies. It aims at redevelop ~15 major colonies and stations into self-sustaining complexes while monetizing assets worth ~INR 2.25 lakh crore. This generates long-term EPC and PPP opportunities in civil works, utilities, and electrical infrastructure, including EPC-cum-O&M models.
- **Regional Rapid Transit System (RRTS)**
First-of-its-kind regional high-speed transit network under NCRTC. It aims at Implement Delhi-Meerut,

Delhi–Alwar, and Delhi–Panipat corridors using Namo Bharat trains (up to 180 km/h) to reduce travel time and congestion. Expands EPC scope in viaduct construction, signaling, 25 kV AC electrification, and multimodal station integration, setting a precedent for Tier-1 and Tier-2 networks.

- **Dedicated Freight Corridors (DFC)**

Backbone freight network under the PM Gati Shakti Plan. It aims at complete Western and Eastern DFCs and extend to East–West, North–South, and East–Coast Corridors, with FY26 capex of ~₹500 crore. This boosts demand for track works, signaling, power systems, and logistics hubs, improving freight capacity and operational efficiency.

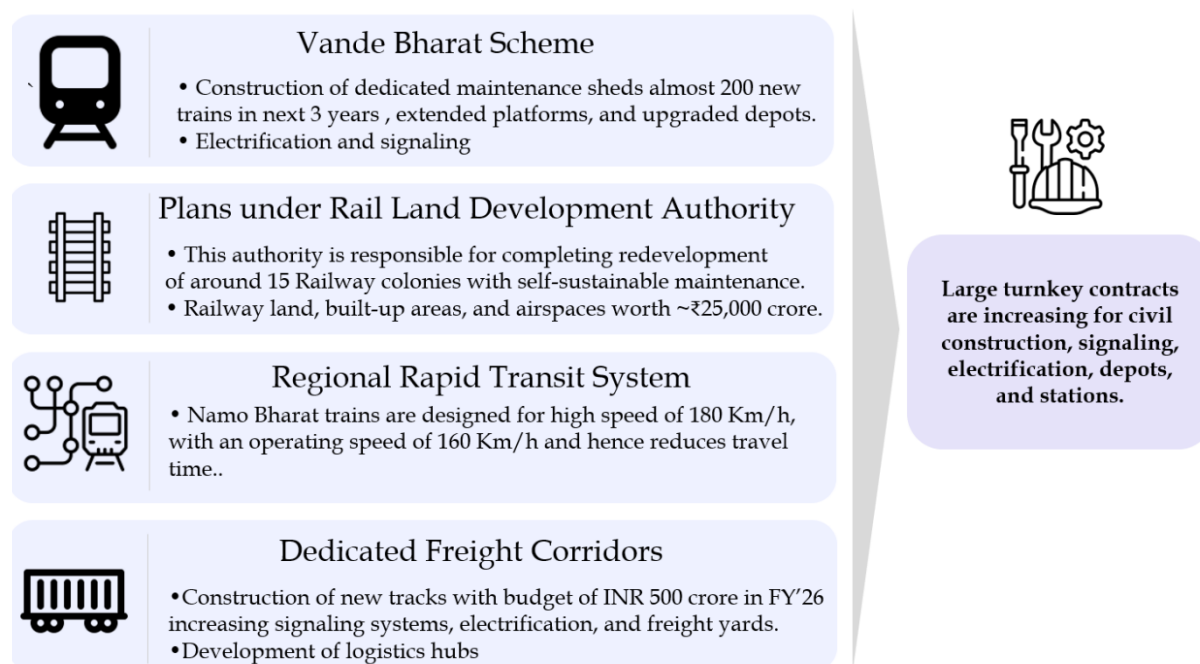
- **Speed Improvement Program (SIP)**

Network-wide modernization initiative for faster passenger train movement. It aims at upgrading 30,000 km of existing lines for 120–160 km/h operations through track renewal, bridge strengthening, automatic signaling (Kavach), and OHE enhancements. It spurs EPC opportunities in civil, signaling, and traction power modernization, with integrated upgrades across major corridors.

- **Freight Tonnage Expansion Plan**

Strategic initiative to scale India's rail freight capacity in line with industrial and logistics growth. It aims at increasing freight volume from 1.6 Bn tonnes (FY24) to 3 Bn tonnes by 2030, using ~3,000 MT-capacity rakes, heavier axle loads, and new goods terminals, GCTs, and multimodal parks. It creates sustained EPC and MEP demand across terminal construction, logistics hubs, traction systems, and yard modernization, integrating with DFCs, ports, and industrial corridors.

Figure 12-1: Major Government Initiatives creating demand for EPC Players



Source: Ken Research Analysis

• **Mission Electrification**

A national initiative by Indian Railways to achieve 100% electrification of broad-gauge routes, enhancing energy efficiency and reducing fuel dependence. Between FY2018 and FY2023, over 58,000 route kilometers were electrified, establishing India as one of the world's largest electrified rail networks. The program drove strong EPC demand in OHE works, traction substations, power distribution, and SCADA integration, modernizing rail infrastructure across key corridors.

• **Station Redevelopment**

A flagship program to redevelop 1,300+ railway stations into modern, passenger-centric, and multi-modal hubs. Aims to enhance commuter experience through upgraded concourses, platforms, circulation areas, and urban integration, implemented under PPP and EPC-cum-O&M frameworks. This creates significant EPC opportunities in civil construction, architectural design, MEP systems, and electrical and signaling integration,

transforming railway stations into integrated transport and commercial centers.

- **DDMC & High-Speed Rapid Railways:**

Opportunities for EPC players lie in developing regional and high-speed corridors enhancing urban-rural connectivity. These projects drive demand for civil works, track laying, electrification, and signaling. As India expands intercity and semi-high-speed routes, EPC scope grows across design, construction, and integrated system works.

Impact of Regional Tendering on Market Accessibility and Execution

The shift to regional tendering in railway EPC has redefined how projects are awarded and executed. By empowering zonal authorities to issue tenders, the system has opened doors for firms with strong local presence and operational agility. This decentralization has led to a more competitive environment, where participation is no longer limited to large national players.

A key outcome of this model is the acceleration of project timelines. Zonal fast-track mechanisms have reduced tender processing durations from 2-3 years to just 6-9 months. This has enabled quicker mobilization, faster ground execution, and improved alignment with regional infrastructure priorities.

As a result, execution capabilities have strengthened, and the overall efficiency of the railway infrastructure ecosystem has improved.

12.2. INITIATIVES DRIVING DEMAND FOR EPC PLAYERS UNDER DRINKING WATER AND WASTE WATER TREATMENT MARKET

India's efforts to strengthen water security are creating new work for engineering, procurement, and construction companies through three major government programs. The **Jal Jeevan Mission**, extended to 2028, is modernizing treatment plants, using advanced technologies, and ensuring long-term operations in rural areas. The **Atal Mission for Rejuvenation and Urban Transformation 2.0** is expanding urban water and sewage networks across 500 cities with a focus on innovation and efficiency. The **Namami Gange Mission**, with an investment of thirty thousand crore rupees, is rejuvenating the Ganga River basin through advanced treatment plants and green financing. **Together, these programs combine scale, technology, and sustainability, placing engineering, procurement, and construction firms at the front of India's water infrastructure growth.**

AMRUT & AMRUT 2.0

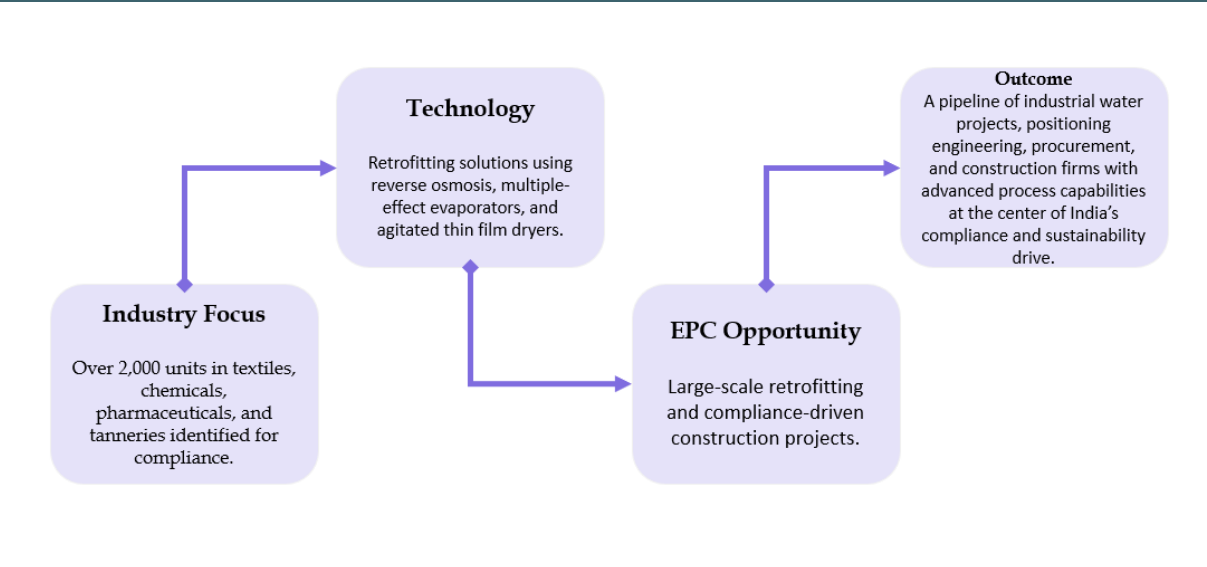
Launched in **2015** and expanded in **2021**, AMRUT and AMRUT 2.0 aim to transform urban water infrastructure across 500+ cities. The combined missions focus on universal coverage of tap and sewer connections, rejuvenation of water bodies, and smart monitoring through SCADA systems. With a total budget of **INR 2.99 lakh crore**, they have already delivered over **2 crore tap** and **1.5 crore sewer connections**, while also enhancing the capacity of Urban Local Bodies (Source: India Water Portal). These initiatives are unlocking large-scale EPC opportunities in water supply, treatment, and urban resilience, making them central to India's urban infrastructure strategy.

National Green Tribunal (NGT)

NGT has introduced a regulatory push through its enforcement of Zero Liquid Discharge (ZLD) norms across key industrial sectors such as textiles, chemicals, pharmaceuticals, and tanneries. **Over 2,000 industrial**

units have been identified for compliance upgrades, creating a robust pipeline of **EPC work in industrial clusters** across **Tamil Nadu, Gujarat, and Maharashtra**. EPC firms with expertise in process engineering and industrial automation are particularly well-suited to deliver retrofitting solutions involving technologies like reverse osmosis (RO), multiple-effect evaporators (MEE), and agitated thin film dryers (ATFD). There is also growing interest in bundling operations and maintenance (O&M) contracts to ensure long-term sustainability and revenue continuity.

Figure 12-2: NGT Enforcement Driving Industrial Water EPC Opportunities



Source: Ken Research Analysis

Together, these initiatives represent a tailwind for India’s Water EPC sector. With combined funding and a clear policy thrust toward sustainability, compliance, and infrastructure modernization, EPC firms that can deliver integrated, tech-enabled, and scalable solutions.

Jal Jeevan Mission

Jal Jeevan Mission (launched in 2019) is generating large-scale demand for water treatment plants, distribution

pipelines, and monitoring solutions. The **Union Budget FY25-26**, the government increased the Mission's **outlay to INR 67,000 crore** and extended it to **2028**, targeting 100% coverage within three years (Source: Ministry of Finance). By targeting every rural household with functional tap water, the mission has expanded the market footprint into remote geographies and created steady opportunities for EPC players specializing in low-cost and sustainable water supply systems.

River Restoration Plans

National Mission for clean Ganga is an initiative under the Namami Gange mission, launched in 2014 with the aim of rejuvenating and conserving the Ganga River, with a estimated **budget allocation** of **INR 3,400 crore** in **FY25-FY26** (Source: Ministry of Jal Shakti). It is an integrated conservation mission launched by the Government of India to rejuvenate the Ganga River **through initiatives like sewerage treatment, river-front development, and biodiversity enhancement**. Key components of the mission include river-surface cleaning, afforestation, public awareness, **industrial effluent monitoring**, and the development of Ganga Gram. As this mission, aims to improve water quality and enhance the ecosystem, **plans such as the sewage treatment development and the industrial effluent monitoring are expected to increase the EPC opportunities in the market**.

The Yamuna Action Plan and river restoration projects for rivers such as the Musi, Cooum, and Sabarmati focus on reducing pollution, afforestation, and riverfront development. These measures directly intersect with the water and water treatment sector, as reducing untreated sewage and industrial effluents entering rivers requires extensive deployment of sewage treatment plants, effluent treatment plants, and decentralized treatment systems (Source: Namami Gange Programme).

Smart Cities Mission

The Smart Cities Mission, with an outlay of INR 1.64 lakh crore and nearly 93% project completion, has been a major growth driver for India's drinking water and wastewater treatment market. (Source: Smart Cities Mission) Many projects focus on water supply, smart metering, SCADA-based leakage detection, and sewerage upgrades, **driving sustained EPC demand**. By integrating automation and digital monitoring, the mission has shifted the market toward tech-enabled solutions. As utilities emphasize real-time water quality, analytics, and efficient sewage management, **EPC opportunities are created in advanced STPs, IoT-based monitoring, and decentralized treatment systems**.

12.3. INITIATIVES DRIVING DEMAND FOR TRANSMISSION AND DISTRIBUTION IN INDIA

India's efforts to strengthen its electricity grid and integrate renewable energy are generating consistent opportunities for engineering, procurement, and construction (EPC) companies. Three major government programs define the current policy landscape:

Revamped Distribution Sector Scheme (RDSS):

Launched in 2021 with an investment of over ₹3 trillion, RDSS aims to modernize India's power distribution network through feeder segregation, system strengthening, and smart meter deployment. It links funding to loss reduction and billing efficiency, ensuring continuous EPC demand across substation upgrades and digital infrastructure.

Green Energy Corridor (GEC):

Implemented in two phases, the GEC is building high-voltage transmission corridors to evacuate renewable power from generation zones to demand centers. Covering over 10,000 ckm of intra-state and inter-state lines, it has created strong EPC demand in Rajasthan, Gujarat, and Tamil Nadu.

National Electricity Plan (NEP) 2023–2032:

The NEP outlines the addition of 1.9 lakh circuit kilometers of new transmission lines and major substation capacity expansion to enable 500 GW of renewable integration by 2030. It promotes private participation through tariff-based competitive bidding, widening EPC opportunities at both central and state levels.

Together, these initiatives combine **grid modernization, renewable integration, and digital transformation**, providing a long-term, policy-backed pipeline for India's transmission and distribution EPC sector.

12.4. INITIATIVES DRIVING DEMAND FOR POWER DISTRIBUTION PANELS AND CSS MARKET IN INDIA

Government programs such as the **Revamped Distribution Sector Scheme (RDSS)** and the **Green Energy Corridor (GEC)**—outlined under India's Transmission and Distribution market—also underpin demand for power distribution panels and compact substations by strengthening grid reliability and renewable integration.

Beyond these, **urban development initiatives** have emerged as major growth drivers for electrical and automation systems. The **Smart Cities Mission, AMRUT 2.0, and Make in India** programs are modernizing urban

infrastructure, driving large-scale deployment of **LT/HT panels, compact substations, and intelligent control systems** across commercial, industrial, and residential projects.

These schemes emphasize **energy efficiency, digital monitoring, and space optimization**, creating sustained demand for **modular, automation-ready electrical panels** and compact substations. As cities expand and industries digitalize, panel integrators and EPC firms are increasingly central to delivering reliable, safe, and technology-driven urban power systems.

12.5. INITIATIVES DRIVING DEMAND FOR SCADA AND AUTOMATION INDIA

SCADA and automation systems serve as a common technological layer across all the infrastructure markets discussed above. They are integral to **railway electrification and signaling, water and wastewater treatment, power transmission and distribution, and urban electrical networks**. As India's infrastructure shifts toward digital, efficient, and remotely managed operations, automation is no longer a standalone segment but a **core enabler embedded within EPC and O&M scopes** across sectors. This is enhancing reliability, safety, and lifecycle performance.

13. COMPETITION LANDSCAPE

13.1. ECOSYSTEM OF INDIA'S RAILWAY EPC MARKET

The Indian Railway Infrastructure Spend is overseen by the Ministry of Railways (MoR) and the Railway Boards of States and Regions where the work needs to be carried out. Large-scale works are assigned to PSUs like IRCON, RITES, and RVNL, which act as nodal agencies for project preparation, tendering (via GeM, CPPP, PSU portals), quality supervision, and sometimes direct EPC execution.

The market is highly competitive, with an estimated **200+ EPC players** primarily Indian firms, alongside select global participants like Siemens, Alstom, and Hitachi in signalling and electrification.

EPC contractors are broadly split into two categories:

- **Fully integrated EPC players:** deliver end-to-end projects across track laying, electrification, bridges, signaling, and station upgrades, either independently or with PSUs.
- **Domain-specific contractors:** specialize in specific verticals such as track laying, electrification, or signalling and telecommunication. **These firms focus on PSU or integrated EPC tenders, often with higher efficiency in their niche.**

Players are further classified by **annual revenue sizing bands**. This indicates the project tender value that the players can achieve in the Railway EPC market.

13.2. ECOSYSTEM OF INDIA'S DRINKING WATER AND WASTEWATER TREATMENT MARKET

In the Indian drinking water and wastewater infrastructure sector large EPC contracts are tendered through state procurement portals and in cases of industrial plants EPC contracts are often negotiated. And localized packages of the large EPC contracts are often bid to regional specialists.

On basis of this, the ecosystem of players in India's drinking water and wastewater market can broadly be classified into three categories:

- **Turnkey EPC Solution Providers:** These companies deliver the full scope of work from design and engineering to procurement, installation, and commissioning of treatment plants.
- **Technology Providers:** These companies provide proprietary treatment equipment, modules, and technologies that are integrated into projects by EPC contractors.
- **Focused EPC & Service Providers:** These are smaller players that execute specific packages within larger projects, such as **civil works, electrical panels, SCADA & automation, or modular STPs/ETPs.**

13.3. ECOSYSTEM OF INDIA'S TRANSMISSION AND DISTRIBUTION MARKET

The Indian Transmission and Distribution (T&D) market covers extra-high-voltage transmission systems and medium- to low-voltage distribution networks. Projects are implemented under central and state programs such as **RDSS, Green Energy Corridor, and Tariff-Based Competitive Bidding (TBCB)**. The market is **moderately fragmented**, with about **250–300 active players** across

transmission, distribution, and equipment supply. Execution responsibility is divided between public sector undertakings, large EPC firms, and regional contractors. Ecosystem participants can be grouped into two categories:

- **Turnkey EPC Players:** Execute complete projects involving design, procurement, installation, and commissioning. These firms typically handle **132-765 kV** lines, substations, and renewable evacuation systems.
- **Subcontracting and Regional Execution Players:** Operate mainly in **11-220 kV** sub-transmission and distribution works under **state DISCOMs** and **RDSS** programs. They execute feeder segregation, substation upgrades, cabling, and metering projects, often as subcontractors to larger EPCs.

13.4. ECOSYSTEM OF INDIA'S POWER DISTRIBUTION PANELS AND COMPACT SUBSTATION MARKET

The Indian power distribution panel and compact substation market serves industrial, commercial, and utility customers. There are players who **assemble complete panels and compact substations**, delivering ready-to-install solutions. Others **manufacture specific components** such as breakers, switches, relays, and busbars that are integrated into these assemblies.

On this basis, the ecosystem of players in India's market can broadly be classified into two categories:

- **Complete Assembly Providers:** These companies deliver fully assembled HT/LT panels or compact substations.
- **Component Suppliers:** These companies manufacture and supply individual electrical components that are integrated into assemblies.

13.5. ECOSYSTEM OF INDIA'S SCADA AND AUTOMATION MARKET

The Indian SCADA and automation market spans across power, water, oil & gas, transport, and industrial applications. Large-scale projects are typically embedded within EPC contracts or smart city/state utility programs, with execution led by a mix of multinational firms and regional specialists. The market is estimated to have 120–150 active players, ranging from global automation leaders to localized integrators.

Ecosystem participants can be broadly split into two categories:

- **Turnkey SCADA & Automation Integrators:** Deliver end-to-end solutions covering design, integration, installation, and commissioning of automation systems.
- **Technology & Platform Providers:** Supply proprietary SCADA software, PLCs, RTUs, sensors, and industrial IoT platforms that are integrated into projects by system integrators

13.6. OVERVIEW OF MAJOR PLAYERS IN INDIA'S INFRASTRUCTURE EPC MARKET

This section includes a cross-comparison between peers operating in India's Infrastructure EPC Market.

Company Profiles:

Core4 Engineers Limited: Incorporated on 2022, the company operates as an electrical-infrastructure and EPC-focused company. The company derives majority of its revenue from water and wastewater treatment, and railway EPC works. It also operates in the transmission and panel segment. The product and solution portfolio spans HT panels, LT distribution boards, power & control centers motor control centers (MCC), compact substations, and SCADA-linked electrical solutions, serving utility and infrastructure project requirements.

Trimurthi HITECH Co. Pvt. Ltd: Incorporated in 1989, the company operates as an electrification contractor with execution exposure concentrated in Indian Railways. Its revenue model is primarily project-based contracting for railway electrification packages and allied electrical works. The operating portfolio includes overhead electrification and commissioning of substations/switching stations, cabling, and allied electrification works, with project experience referenced for Southern Railway.

SN Enviro-Tech Pvt. Ltd: The company is an EPC service provider in water and wastewater infrastructure space. Its revenue model is anchored in engineering, procurement and construction contracts for public/industrial water systems, including sewage and effluent treatment projects. The portfolio includes EPC delivery of water & sewage treatment plants and industrial effluent treatment plants, alongside

associated civil and systems execution as part of turnkey project scopes.

K2 Infragen Ltd: Incorporated in 2015, it operates as an integrated EPC and core infrastructure player with multi-state execution exposure. Its revenue model is diversified across EPC contract revenues, services/project structuring & execution support, and (as disclosed in offer-note summaries) trading/procurement-led revenues alongside core EPC. The portfolio covers end-to-end execution (design to commissioning) across water supply projects, railway projects, road projects, civil construction works, and related infrastructure builds, with delivery spanning multiple Indian states.

Table 13-1: Cross-Comparison of Peers in India Infrastructure EPC Market on basis of Operational Parameters, as on February 2026

Players	Year Incorporated	Operational Coverage	Product Portfolio
Core4 Engineers Ltd.	2022	Pan India	• Fabrication of Distribution Panels • SCADA Integration • Waste Water Treatment Plants • Railway EPC • compact substation, transmission and distribution
Trimurthi HITECH Co. Pvt. Ltd.	1989	Predominantly Southern India. Notable Presence in Central India too.	• Turnkey Project Execution • Electronic & Instrumentation Works • Civil Construction Works • Mechanical & Piping Works • Electrical Works (Railway and Power)
SN Enviro-Tech Pvt. Ltd.	2005	Across 15 States in India	• Wastewater treatment • Drinking Water Production and Distribution • Seawater Desalination • Wastewater Recycling • Aquifer Recharge • Effluent Treatment

Players	Year Incorporated	Operational Coverage	Product Portfolio
K2 Infragen Ltd.	2015	Present in 9 states in India	EPC services in the domains of Roads & Bridges, Power Transmission & Distribution, Railways, Water Supply Projects and Buildings Construction

Source: Companies' Websites, Annual Reports & Ken Research Analysis

13.7. FINANCIAL SNAPSHOT OF MAJOR PLAYERS

This section includes a financial cross-comparison between peers operating in India's Infrastructure EPC Market.

Table 13-2: Cross-Comparison of Major Key Peers in India Infrastructure EPC Market on basis of Financial Parameters, FY23 - FY25

Company	Financial Year	Revenue From Operations (INR Lakh)	EBITDA (INR Lakh)	EBITDA Margin	PAT (INR Lakh)	PAT Margins
Core4 Engineers Ltd.	FY26 (H1)	4,707.0	909.1	19.3%	605.4	12.9%
	FY25	3,400.1	759.3	22.3%	409.8	12.1%
	FY24	4,010.4	578.4	14.4%	305.3	7.6%
	FY23	1,312.0	183.5	14.0%	94.0	7.2%
Trimurthi HITECH Co. Pvt. Ltd.	FY25	NA	NA	NA	NA	NA
	FY24	7,020.8	288.0	4.1%	125.6	1.8%
	FY23	3,451.0	153.9	4.5%	42.8	1.2%
SN Enviro-Tech Pvt. Ltd.	FY25	NA	NA	NA	NA	NA
	FY24	NA	NA	NA	NA	NA
	FY23	5,107.7	816.8	16.0%	288.9	5.7%
K2 Infragen Ltd.	FY25	14,680.8	2,020.1	13.8%	1,154.3	7.9%
	FY24	10,871.8	2,142.6	19.7%	1,250.2	11.5%
	FY23	7,479.1	1,937.8	25.9%	1,128.0	15.1%

Source: Companies' Websites, Annual Reports, Proprietary Databases & Ken Research Analysis

EBITDA = PBT + Finance cost + Depreciation and Amortization

PAT = PBT - Total Tax Expense

Return on Net Worth (%) ROE = PAT/Average Shareholder's Equity

Capital Employed = Total equity + Deferred Tax Liabilities, long term borrowings and short-term borrowings - Deferred Tax Assets

ROCE (%) = EBIT/Average Capital Employed

ROA = PAT/Average Total Assets

Note 1: FY25 indicates financial year which starts from 1st April 2024 and ends on 31st March 2025

Note 2: N/A indicates not available; annual filling not reported to MCA.

Note 3: FY26 (H1)- For ROC, ROE, and ROA, the figures have been annualized.

Note 4: FY 23 figures represent standalone financials for Core4 Engineers Ltd., while FY 24, FY 25 and FY 26 (H1) figures are presented on a consolidated basis

Table 13-3: Cross-Comparison of Major Key Peers in India Infrastructure EPC Market on basis of Financial Parameters, FY23 – FY25

Company	Financial Year	Return on Capital Employed (ROCE)	Return on Assets (ROA)	Return on Net Worth (ROE)
Core4 Engineers Ltd.	FY26 (H1)	42.5%	16.6%	55.0%
	FY25	24.2%	8.1%	31.4%
	FY24	N/A	N/A	N/A
	FY23	N/A	N/A	N/A
Trimurthi HITECH Co. Pvt. Ltd.	FY25	NA	NA	NA
	FY24	11.1%	2.8%	21.2%
	FY23	6.5%	1.1%	7.5%
SN Enviro-Tech Pvt. Ltd.	FY25	NA	NA	NA
	FY24	NA	NA	NA
	FY23	9.4%	2.7%	6.0%
K2 Infragen Ltd.	FY25	13.9%	8.1%	19.0%
	FY24	28.3%	15.8%	42.3%
	FY23	47.1%	25.9%	107.2%

Source: Companies' Websites, Annual Reports, Proprietary Databases & Ken Research Analysis

EBITDA = PBT + Finance cost + Depreciation and Amortization

PAT = PBT - Total Tax Expense

Return on Net Worth (%) ROE = PAT/Average Shareholder's Equity

Capital Employed = Total equity + Deferred Tax Liabilities, long term borrowings and short-term borrowings - Deferred Tax Assets.

ROCE (%) = EBIT/Average Capital Employed

ROA = PAT/Average Total Assets

Note 1: FY25 indicates financial year which starts from 1st April 2024 and ends on 31st March 2025

Note 2: N/A indicates not available; annual filling not reported to MCA.

Note 3: FY26 (H1)- For ROC, ROE, and ROA, the figures have been annualized.

Note 4: FY 23 figures represent standalone financials for Core4 Engineers Ltd., while FY 24, FY 25 and FY 26 (H1) figures are presented on a consolidated basis

14. RESEARCH METHODOLOGY

14.1. ABBREVIATIONS

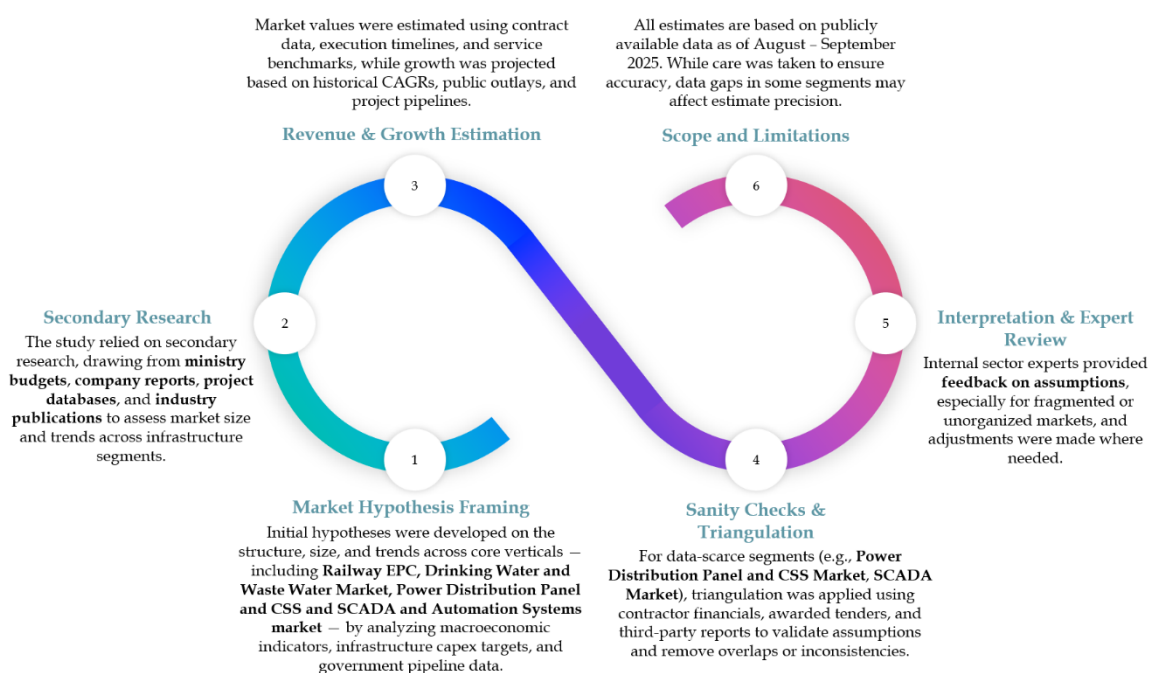
ABSS	Amrit Bharat Station Scheme
AI	Artificial Intelligence
AMC	Annual Maintenance Contract
AMRUT 2.0	Atal Mission for Rejuvenation and Urban Transformation 2.0
AOPs	Advanced Oxidation Processes
AT&C	Aggregate Technical & Commercial
ATFD	Agitated Thin Film Dryer
BIS	Bureau of Indian Standards
BOD	Biochemical Oxygen Demand
BOQ	Bill of Quantities
CAPEX	Capital Expenditure
CAGR	Compound Annual Growth Rate
CEA	Central Electricity Authority
CERC	Central Electricity Regulatory Commission
CGWA	Central Ground Water Authority
CKM	Circuit Kilometer
CPCB	Central Pollution Control Board
CPPP	Central Public Procurement Portal
CPRI	Central Power Research Institute
CSS	Compact Substation
CY	Calendar Year
DFC	Dedicated Freight Corridor
DG	Diesel Generator
DMS	Distribution Management System
DPR	Detailed Project Report
DWTPs	Drinking Water Treatment Plants
EPC	Engineering, Procurement, and Construction
ETP	Effluent Treatment Plant
EV	Electric Vehicle
FAME-II	Faster Adoption and Manufacturing of (Hybrid &) Electric Vehicles in India - Phase II
FAT	Factory Acceptance Tests

FY – Fiscal Year
GCC – General Conditions of Contract
GeM – Government e-Marketplace
GIS – Gas-Insulated Switchgear
GVA – Gross Value Added
HMI – Human-Machine Interface
HT – High Tension
HVDC – High Voltage Direct Current
I/O – Input/Output
ICCCs – Integrated Command and Control Centres
IEC – International Electrotechnical Commission
INR – Indian Rupee
IoT – Internet of Things
IRCON – Indian Railway Construction (IRCON International Limited)
IS – Indian Standard(s)
IT – Information Technology
JNPT – Jawaharlal Nehru Port Trust
kV – Kilovolt
LT – Low Tension
LV – Low Voltage
MBRs – Membrane Bioreactors
MEE – Multiple Effect Evaporators
MEP – Mechanical, Electrical & Plumbing
MoF – Ministry of Finance (publisher of the *Manual for Procurement of Works*)
MoEF&CC – Ministry of Environment, Forest and Climate Change
MoP – Ministry of Power
MT – Metric Tonne
NABL – National Accreditation Board for Testing and Calibration Laboratories
NGT – National Green Tribunal
NHSRCL – National High Speed Rail Corporation Limited
NOCs – No Objection Certificates
NSGM – National Smart Grid Mission
O&M – Operation and Maintenance
OHE – Overhead Electrification
OPEX – Operating Expenditure
OT – Operational Technology
PAT – Profit After Tax

PEB – Pre-Engineered Building
pH – Potential of Hydrogen
PLC – Programmable Logic Controller
PM – Pradhan Mantri (Prime Minister)
PPAC – Public Private Partnership Appraisal Committee (often styled PPPAC)
PPP – Public Private Partnership
PSU/PSUs – Public Sector Undertaking(s)
RDSS – Revamped Distribution Sector Scheme
RFCTLARR – Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement
rites – Rail India Technical and Economic Service
RLDA – Rail Land Development Authority
RO – Reverse Osmosis
ROA – Return on Assets
ROCE – Return on Capital Employed
RRTS – Regional Rapid Transit System
RTU – Remote Terminal Unit
RVNL – Rail Vikas Nigam Limited
SAT – Site Acceptance Tests
SCADA – Supervisory Control and Data Acquisition
SECI – Solar Energy Corporation of India
SoP – Standard Operating Procedure
SPCB – State Pollution Control Board
S&T – Signaling and Telecommunication
STP – Sewage Treatment Plant
T&D – Transmission and Distribution
USBRL – Udhampur-Srinagar-Baramulla Rail Link
VFD – Variable Frequency Drive
WTP – Water Treatment Plant
YoY – Year on Year

14.2. MARKET SIZING AND MODELING

CONSOLIDATED RESEARCH APPROACH



This study relies on **Secondary Research and Expert Opinion Methodology**, drawing from a wide range of publicly available and credible sources to estimate market sizes and understand sectoral dynamics. Data was collected from government publications, ministry reports, infrastructure project databases, annual budget documents, and industry white papers. These sources provided detailed insights into contract awards, project execution timelines, and regional distribution of infrastructure activities.

Additional information was gathered from company websites, investor presentations, and trade association reports to understand market participation, service offerings, and competitive landscape. This approach enabled a comprehensive understanding of market trends, demand drivers, regional variations, and operational models across the four segments.

The use of structured and validated secondary data ensures that the findings are aligned with actual **supply-side activity** and reflect the evolving nature of India's engineering and infrastructure ecosystem.

Further for validations, **Expert Opinion methodology** was carried out for which multiple interviews with industry stakeholders was done, this included:

Table 14-1: Sample Composition Table by Stakeholders and Respondents in (%)

By Stakeholders	Sample Size: ~40 Respondents	Description
Railway EPC Players	25%	Project Managers, Civil/Electrical/Mechanical Engineers, Design Engineers (CAD/CAM/CAE), Planning & Scheduling Heads, Quality Assurance/Control Managers, Site In-Charge / Construction Managers, Business Development Heads, P
Water Treatment Plant Owners and EPC Players	30%	Directors, CXOs, Project Heads, Sales Heads, Managers
Power Distribution Panel and Compact Substation Manufacturers	15%	Directors, CXOs, Business Heads, Sales Managers, Managers
SCADA and Automation System Providers	15%	Procurement Heads, Operations Heads, Project Managers
Industry Experts & Associations	15%	Independent Tooling Consultants, Industry Analysts, Academicians, Leaders from Industrial Associations

Source: Ken Research Analysis

LIMITATIONS

Due to the fragmented nature of India's infrastructure EPC markets, comprehensive datasets on contract values, vendor participation, and regional execution were not consistently available in the public domain. Accordingly, market sizing and segmentation have been developed using proxy indicators such as tender awards, budget allocations, and reported project execution values, which were validated through triangulation with multiple secondary sources.

While every effort has been made to ensure robust representation across segments (*Railway EPC, Drinking Water and Waste Water Treatment, Transmission and Distribution, Power Distribution Panels and Compact Sub-Stations and SCADA and Automation Systems Market*), gaps in public reporting and the prevalence of subcontracting arrangements may result in minor estimation variances. The projections presented are grounded in structured data and sector trends but may not

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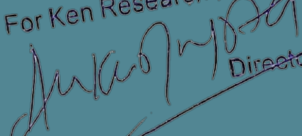
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